

Amazon Bedrock Agent: -

Summary: -

I created a smart assistant that can read your documents, answer questions, and even do simple tasks like telling the time.

It understands what you ask, finds the right information from your files, and gives clear answers. It can also perform small actions when needed.

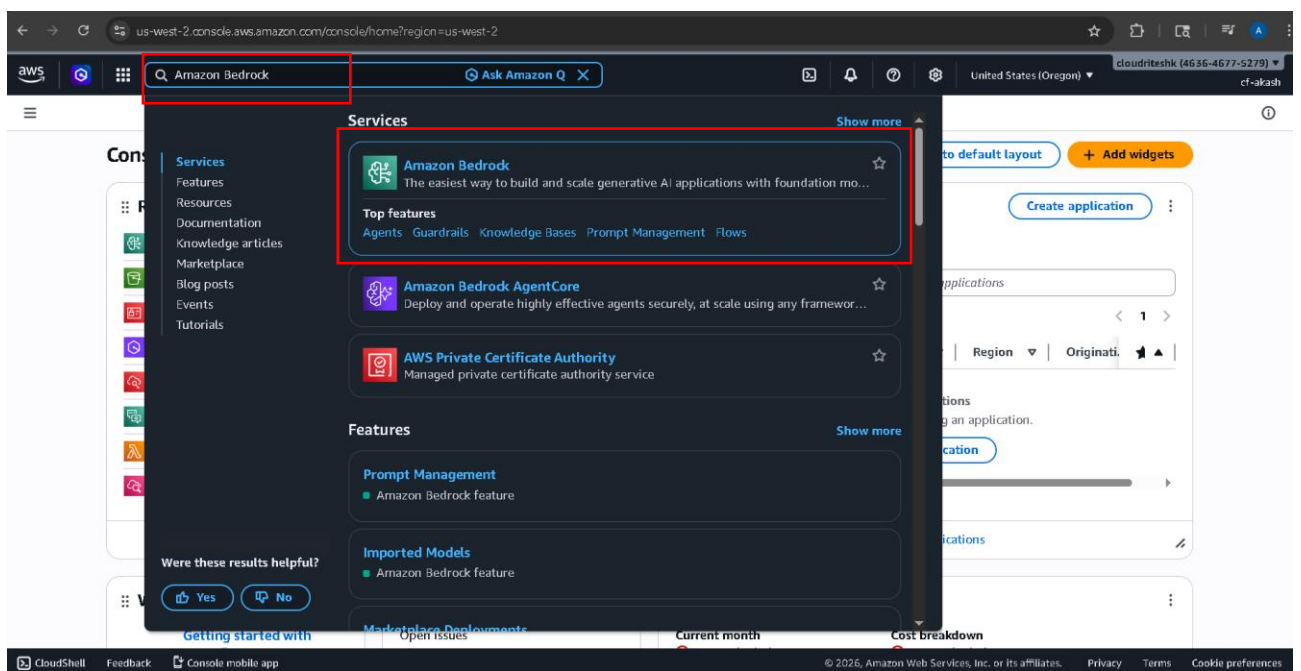
In short: I built a helpful assistant that can answer questions and do basic tasks automatically using your data.

Amazon Bedrock Agent: -

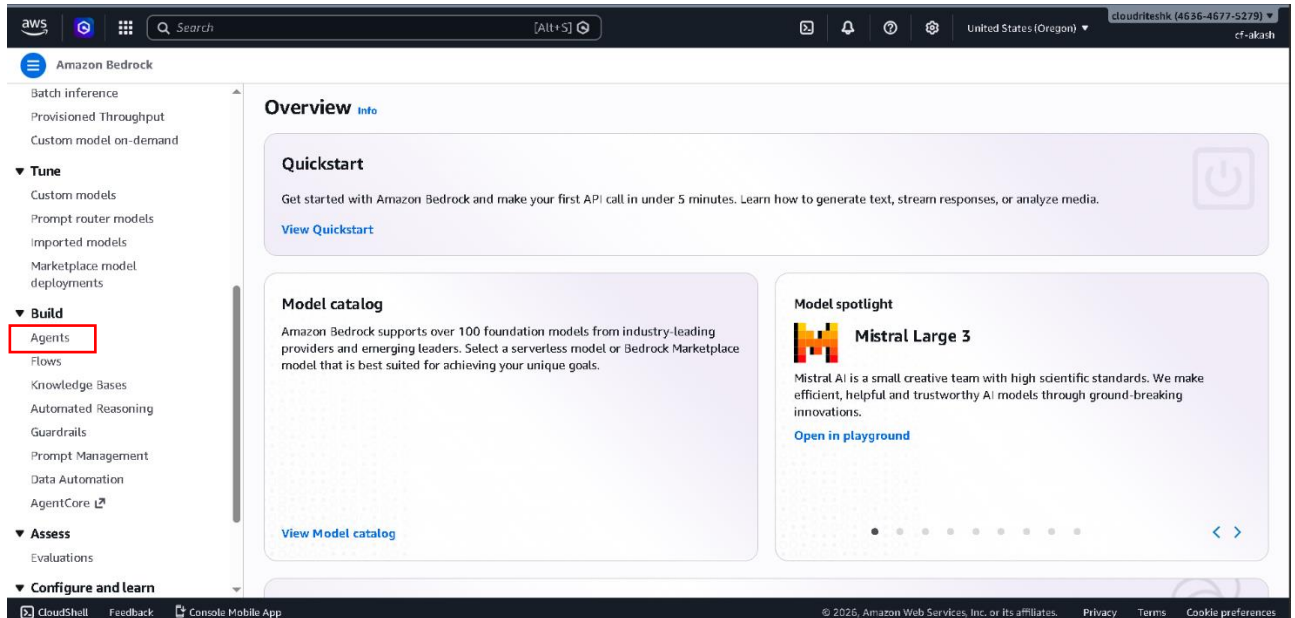
Amazon Bedrock Agents are AI-powered assistants that use foundation models to understand user requests, plan actions, and interact with APIs or data sources to complete tasks automatically.

Now we create our First Bedrock Agent: -

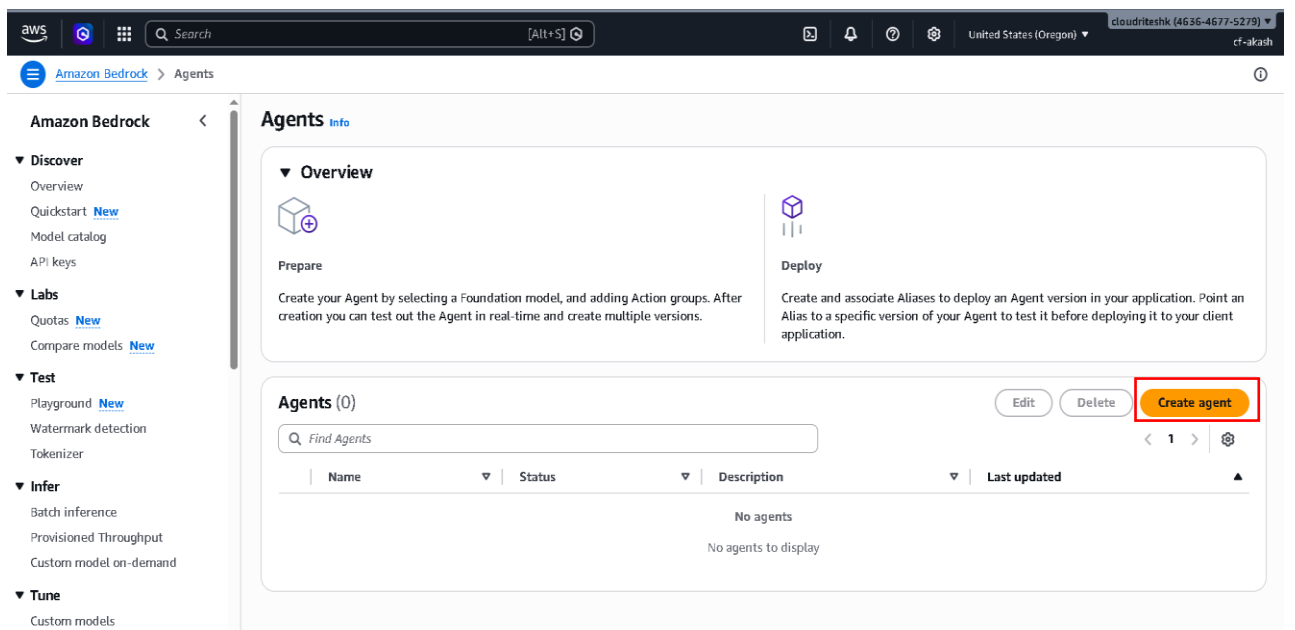
- Go to your AWS console.



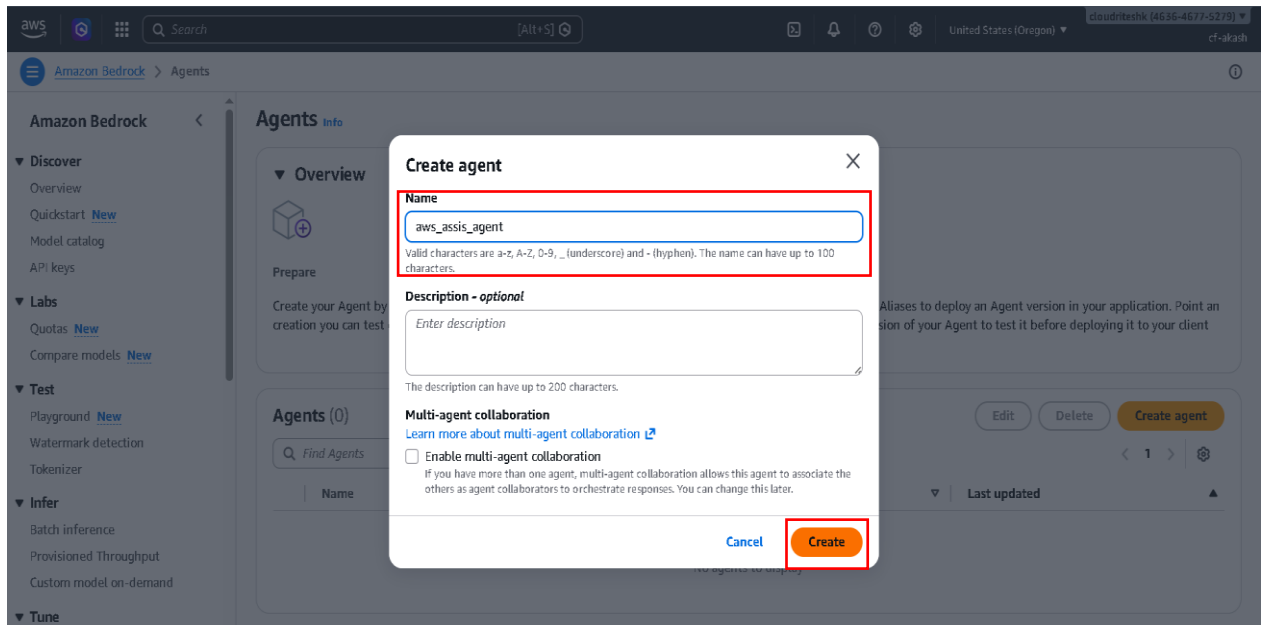
- Click on Search Box and Search Amazon Bedrock.
- Now click on Amazon Bedrock.
- So they redirected to Amazon Bedrock Page.



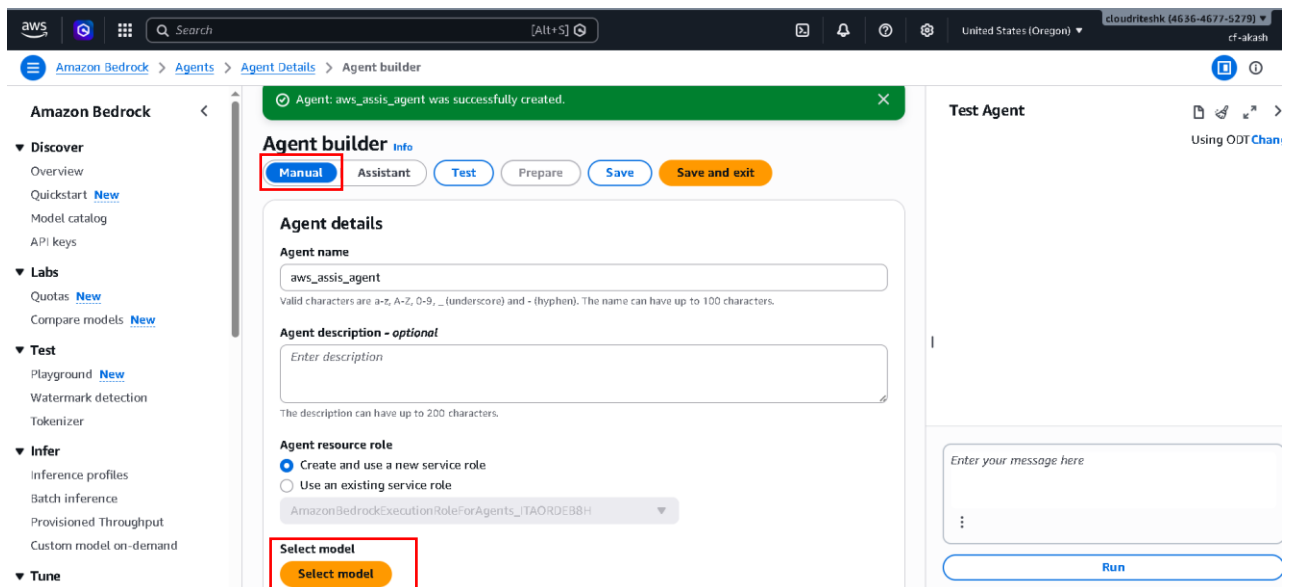
- Now you are in Amazon Bedrock Page.
- Scroll down in navigation bar.
- And now just click Agents in Build Section.



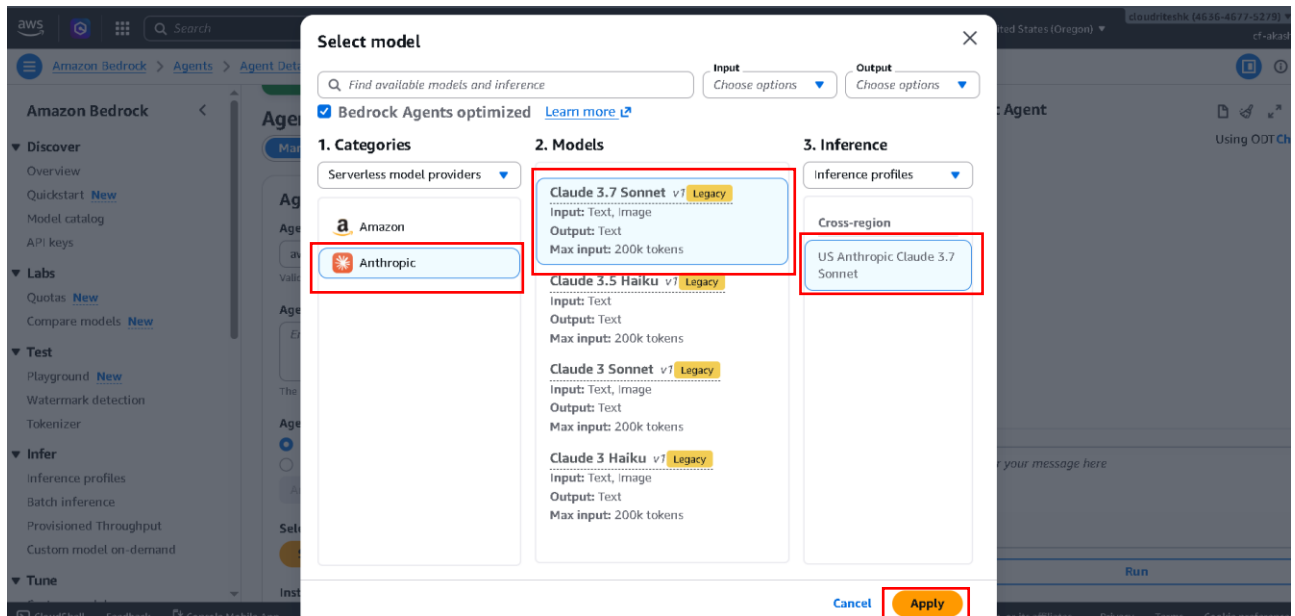
- Now you are in Amazon Bedrock Agent page.
- Now just click on create Agent.



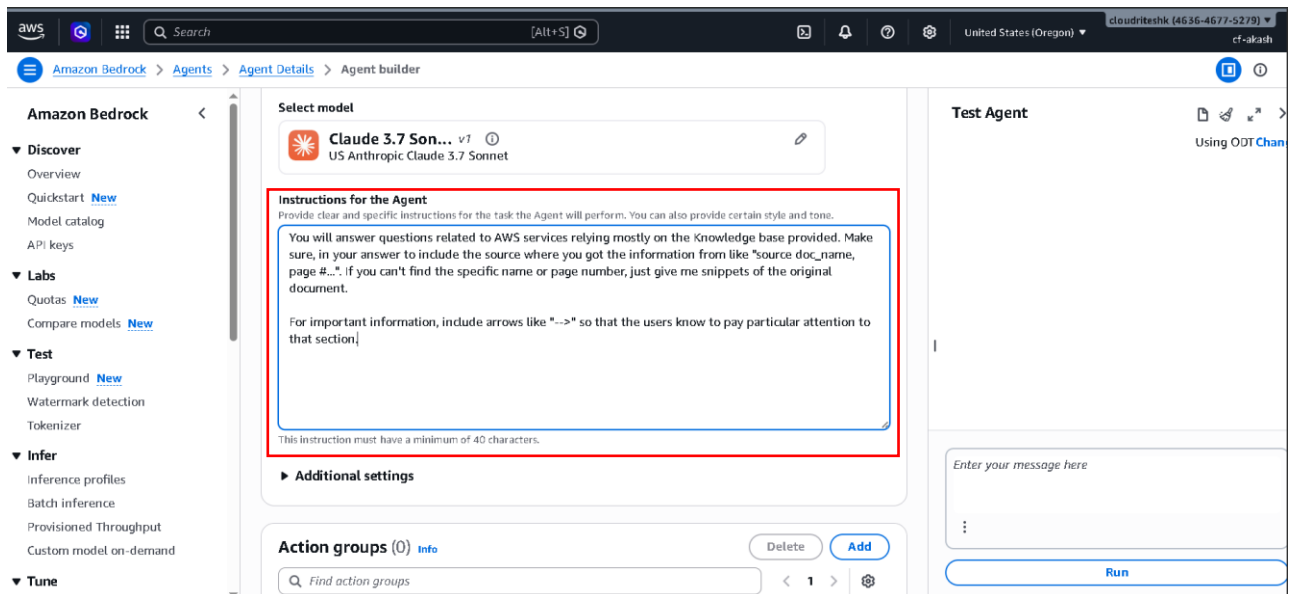
- Give name to your agent.
- And just click Create.



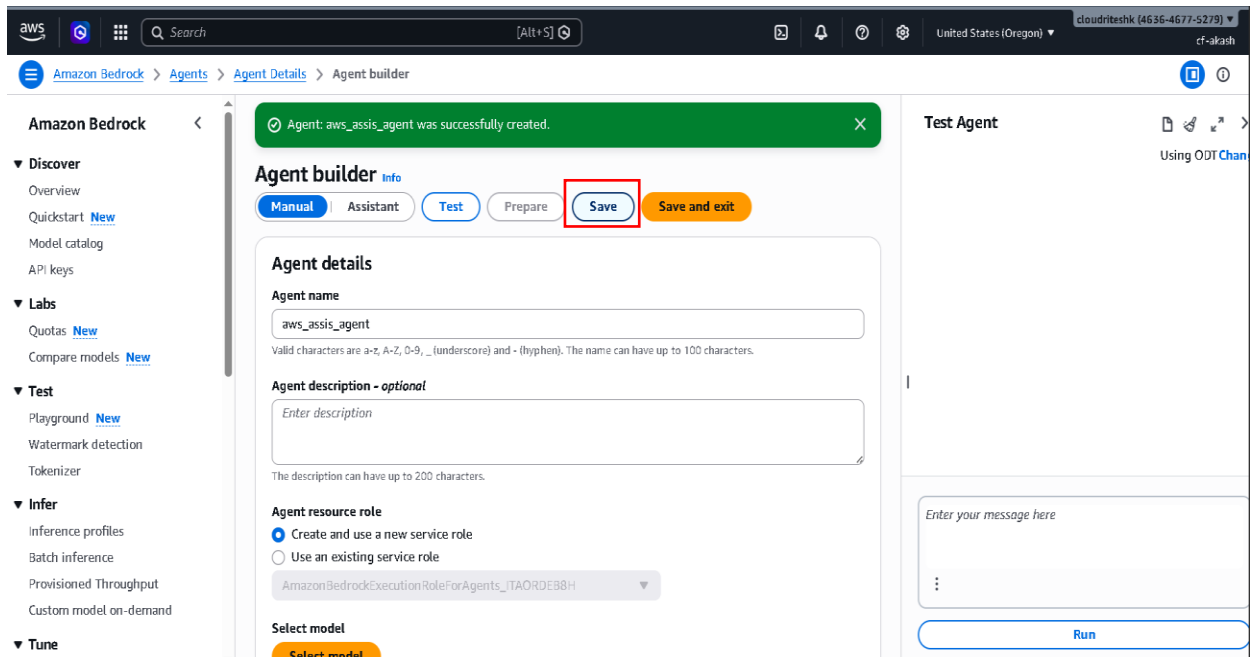
- Our Agent is created.
- Now select Manual.
- And click Select Model.



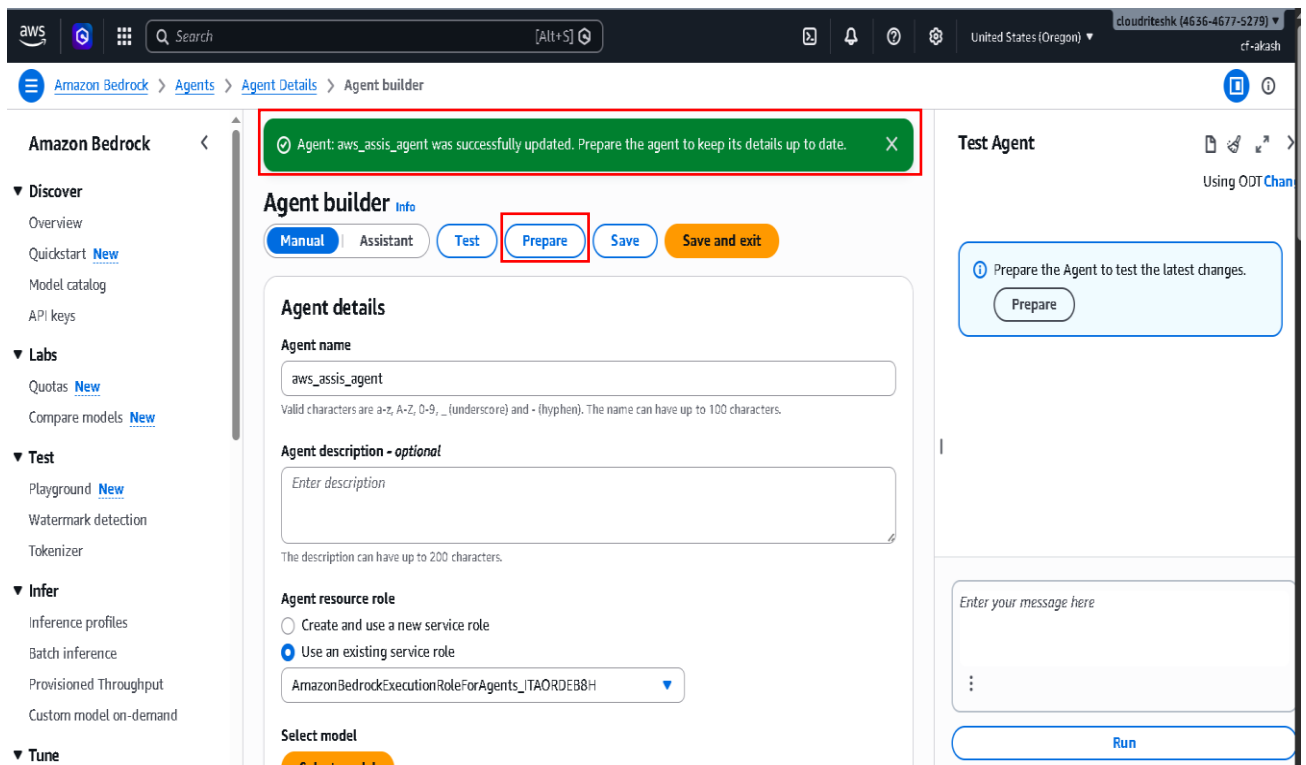
- Select Categories, Model and inference.
- Click on Apply.



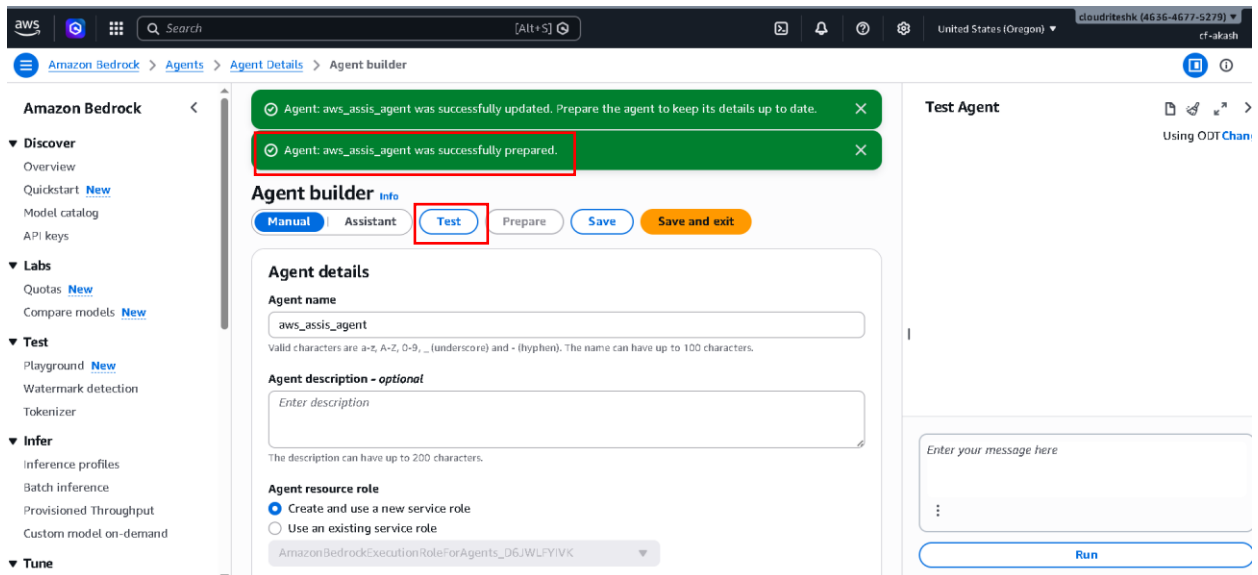
- Now give instruction for the Agent.



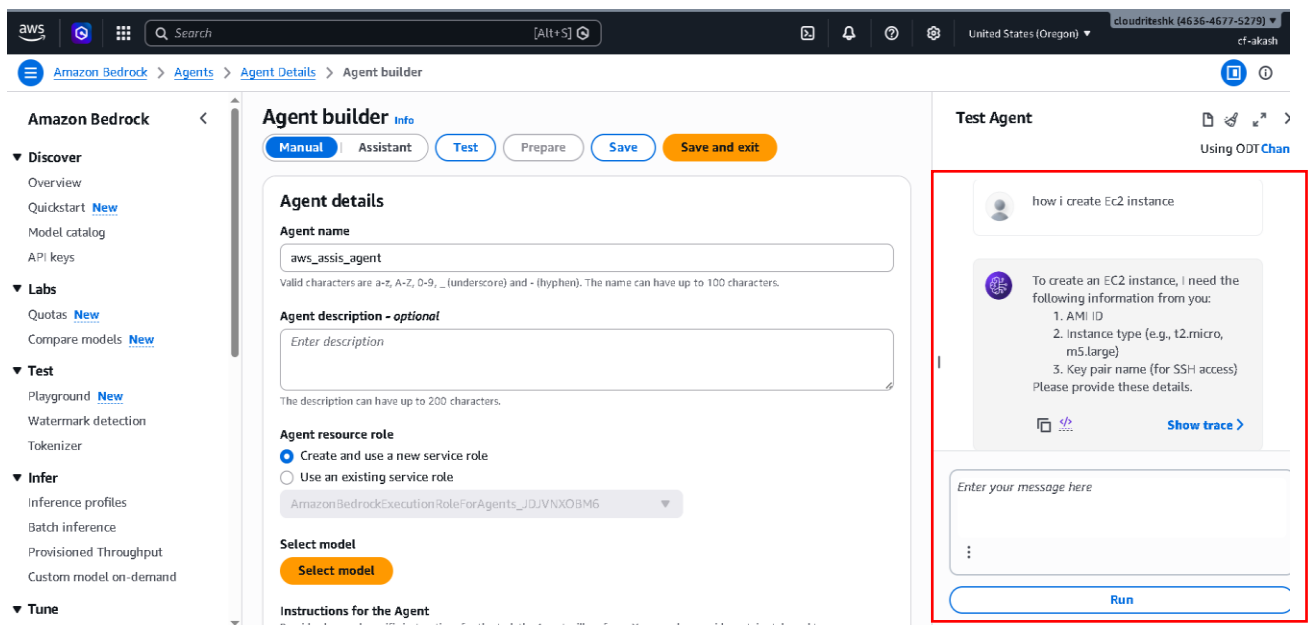
- Now scroll up and select Save.



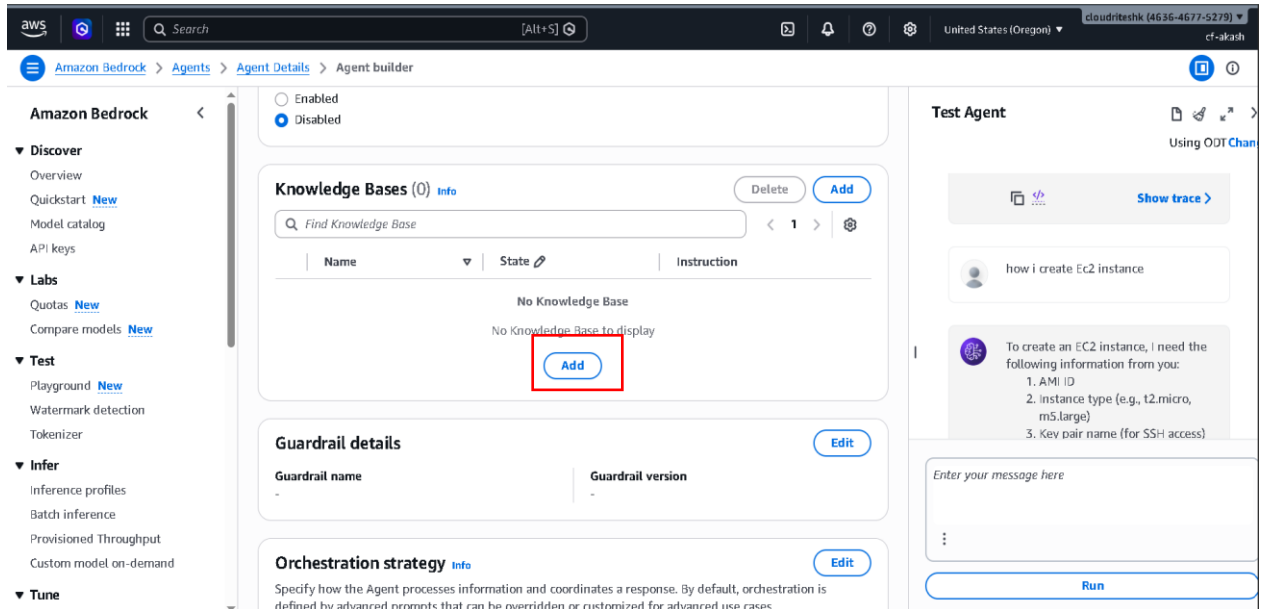
- Our agent is successfully updated.
- Now click Prepare.



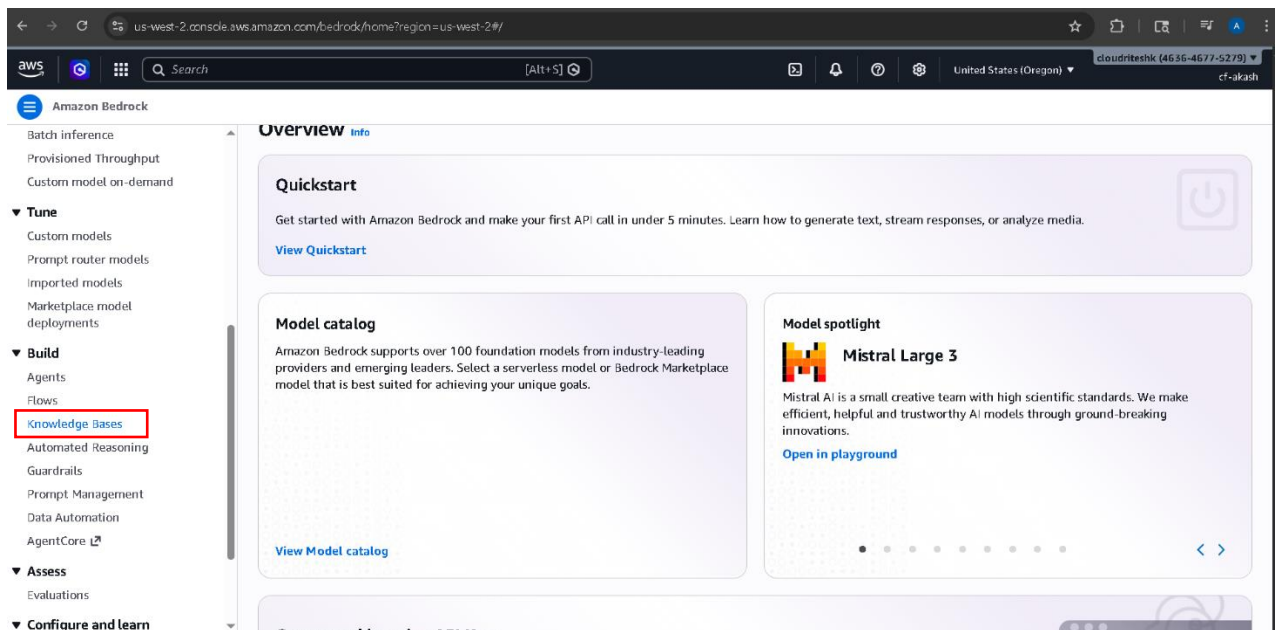
- Our Agent is Prepared.
- Click Test



- Now our agent work properly.
- So now we have a basic Agent.
- So now we add Knowledge base to our Agent.
- So Let's start.



- Scroll Down in Agent Builder
- And now just click Add inside Knowledge Bases.
- Before Adding Knowledge Base we need to create first
- So open Aws in new tab and go to Amazon Bedrock home.



- Scroll down in Navigation bar.
- And select Knowledge Base inside Build.

Amazon Bedrock > **Knowledge Bases** > Create knowledge base with vector store

Step 1: **Provide Knowledge Base details**

Knowledge Base details

Knowledge Base name

 Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 50 characters.

Knowledge Base description - optional

 Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 200 characters.

IAM permissions
 Certain permissions are necessary to access other services or perform actions in order to create this resource. For more information, see [service role](#) for Amazon Bedrock.

Runtime role

☒ Create and use a new service role
☐ Use an existing service role

- Give Name to your Knowledge Base
- IAM permissions
 - Create and use a new service role

Amazon Bedrock > **Knowledge Bases** > Create knowledge base with vector store

AmazonBedrockExecutionRoleForKnowledgeBase_7oqju

Choose data source type
 Select the data source type that you want to configure in the next step. You may add additional data sources once your Knowledge Base is created.

Amazon S3 (selected)
 Object storage service that stores data as objects within general purpose buckets. Add up to 5 buckets as data sources.

Confluence - Preview
 Collaborative work-management tool designed for project planning, software development and product management.

Custom
 Create a data source directly in Amazon Bedrock. A custom data source allows the flexibility to automatically ingest documents into your vector database directly.

Salesforce - Preview
 Customer relationship management (CRM) tool for managing support, sales, and marketing data.

Sharepoint - Preview
 Collaborative web-based service for working on documents, web pages, web sites, lists, and more.

Web Crawler - Preview
 Web page crawler that extracts content from public web pages you are authorized to crawl.

- Choose data Source Amazon S3.
- And simply scroll down and click Next.

Step 2
 Configure data source

Step 3
 Configure data storage and processing

Step 4
 Review and create

Configure for the chosen data source

Amazon S3
 Provide details to connect Amazon Bedrock to your S3 data source.

▼ Data source: aws_agent_assist

Data source name
 aws_agent_assist

Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 100 characters.

Data source location
☒ This AWS account
☐ Other AWS account

S3 URI [Info](#)
 Enter the S3 bucket location containing your source files. To enhance search relevance and filtering capabilities, you can include metadata by adding a .metadata.json file for each source file.

s3://bucket-name/prefix/object [View](#) [Browse S3](#)

☐ KMS Key for S3 data - optional
 If you encrypted your S3 data, provide the KMS key here so that Bedrock can decrypt it.

- Now Give data source name
- Now for S3 URI we Need S3 Bucket.
- So open Aws console in new tab and search S3 Bucket.

Amazon S3 > Buckets

General purpose buckets [All AWS Regions](#) Directory buckets

General purpose buckets (133) [Info](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	Creation date
ab-aws-de-labs	Asia Pacific (Mumbai) ap-south-1	February 12, 2026, 00:44:57 (UTC+05:30)
amazon-cdimagesdata	Asia Pacific (Mumbai) ap-south-1	November 13, 2025, 15:18:01 (UTC+05:30)

Account snapshot [Info](#) [View dashboard](#)
 Updated daily
 Storage Lens provides visibility into storage usage and activity trends.

External access summary [Info](#)
 Updated daily
 External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

- Now just click Create Bucket.

Amazon S3 > Buckets > Create bucket

Bucket type [Info](#)
☒ General purpose
 Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ Directory
 Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
 Choose the namespace where you want to create your bucket. [Learn more](#)
☒ Global namespace
 By default, S3 creates general purpose buckets in the global namespace.

☐ Account Regional namespace (recommended)
 General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)
 aws_assist_file

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
 Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

- Give Bucket Name
- Else keep things default

Default encryption [info](#)
Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [info](#)
Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing on the Storage tab of the Amazon S3 pricing page](#).

- ☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
- ☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
- ☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

- ☐ Disable
- ☒ Enable

► **Advanced settings**

After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

[Cancel](#) [Create bucket](#)

- Now just scroll down and click Create Bucket.

aws-assist-file [info](#)

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [File systems - new](#) [Access Points](#)

Objects (0) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
No objects You don't have any objects in this bucket.				

[Upload](#)

- Our Bucket is ready.
- Go Back to your Bucket.

Buckets

[General purpose buckets](#) [All AWS Regions](#) [Directory buckets](#)

General purpose buckets (1/134) [info](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

aws- 13 matches

Name	AWS Region	Creation date
ab-aws-de-labs	Asia Pacific (Mumbai) ap-south-1	February 12, 2026, 00:44:57 (UTC+05:30)
aws-assist-file	US West (Oregon) us-west-2	April 18, 2026, 12:53:45 (UTC+05:30)
aws-calculator-project1	Asia Pacific (Mumbai) ap-south-1	November 22, 2025, 12:05:49 (UTC+05:30)
aws-cloudtrail-logs-463646775279-046a6e88	Asia Pacific (Mumbai) ap-south-1	November 27, 2025, 18:48:56 (UTC+05:30)
aws-cloudtrail-logs-463646775279-acaf3794	Asia Pacific (Mumbai) ap-south-1	April 9, 2026, 18:36:55 (UTC+05:30)
aws-cloudtrail-logs-463646775279-d86f1d6a	Asia Pacific (Mumbai) ap-south-1	November 20, 2025, 13:13:13 (UTC+05:30)
aws-glue-assets-463646775279-ap-south-1	Asia Pacific (Mumbai) ap-south-1	December 24, 2025, 06:55:23 (UTC+05:30)

Account snapshot [info](#) [View dashboard](#)
Updated daily
Storage Lens provides visibility into storage usage and activity trends.

External access summary [info](#)
Updated daily
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

- aws

Search

[Alt+S] Ask Amazon Q

United States (Oregon) cloudfirehk (4636-677-5279) cf-aksh

Amazon Bedrock > Knowledge Bases > Create knowledge base with vector store

Review and create

Data source name

aws_agent_assist

Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 100 characters.

Data source location

This AWS account

Other AWS account

S3 URI Info

Enter the S3 bucket location containing your source files. To enhance search relevance and filtering capabilities, you can include metadata by adding a .metadata.json file for each source file.

s3://bucket-name/prefix/object

View

Browse S3

KMS Key for S3 data - optional

If you encrypted your S3 data, provide the KMS key here so that Bedrock can decrypt it.

Parsing strategy Info

The Amazon Bedrock Default parser handles common text formats, while specialized parsers are needed for other content types. Any remaining non-parseable content is processed according to the chosen embedding model's capabilities.

Amazon Bedrock default parser

Best if only parsing text. This parser will ignore multimodal content. Commonly used with multimodal embedding model.

Data parsed:
 - Text documents(e.g. Word, Excel, html, markdown, .txt, .csv)

Amazon Bedrock Data Automation as parser

Allows you to parse text and store multimodal content as text. Use this for high accuracy speech retrieval on audio/video files.

Data parsed:
 - PDFs
 - Images
 - Audio

Foundation models as a parser

Optimal for more advanced text and image parsing use cases. You can use the default parser prompt or customize it for your use case.

Data parsed:
 - PDFs
 - images

- Search

[Alt+S] Ask Amazon Q

United States (Oregon)

cloudniteshk (4636-4677-5279)

cf-akash

Amazon Bedrock > Knowledge Bases > Create knowledge base with vector store

Parser output:

 - Extracted text

Parser output:

 - Audio
 - Video

Parser output:

 - Images
 - Structured/formatted text documents
 - Visually rich documents

① If you need high accuracy speech retrieval on audio/video files, select the Bedrock Data Automation as a parser to convert those modalities to text.

Chunking strategy

Chunking breaks down the text into smaller segments before embedding. The chunking strategy can't be modified after you create the data source.

Default chunking

Automatically splits text into chunks of about 300 tokens in size, by default. If a document is less than or already 300 tokens,...

► Transformation function - optional

► Advanced settings - optional

Cancel

Previous

Next

- 11 | Page

Step 2: Configure data source

Step 3: **Configure data storage and processing**

Step 4: Review and create

Choose an embeddings model to convert the data that you will provide in the next step, and provide details for a vector data store in which Bedrock can store, manage, and update your embeddings. The embeddings model and vector store cannot be changed after creation of Knowledge Base.

Embeddings model
Select an embeddings model to convert your data into an embedding. Your selection may limit vector stores and embedding types that are available. Pricing depends on the model. [Learn more](#)

[Select model](#)

Vector store Info
Let Amazon create a vector store on your behalf or select a previously created store to allow Bedrock to store, update and manage embeddings. You will be billed directly from the vector store provider. [Learn more](#)

Vector store creation method

☒ **Quick create a new vector store - Recommended**
A vector store will be created on your behalf in this AWS account during Knowledge Base creation.

☐ Use an existing vector store
Connect to an existing vector store to store, update, and manage embeddings.

Vector store type - new [Info](#)
A vector store holds the vector embeddings representation of your data. This choice cannot be changed later. Not all vector stores support binary vector embeddings.

[Select a vector store](#)

[Cancel](#) [Previous](#) [Next](#)

- Now click on Select Model.

Select model

Find available models and inference

1. Categories

☒ Serverless model providers

☐ Amazon

☐ Cohere

2. Models

☐ Titan Embeddings G1 - Text v1.2
Embedding model | 8k tokens

☒ Titan Text Embeddings V2 v2.0
Embedding model | 8k tokens

3. Inference

☐ On demand

[Cancel](#) [Apply](#)

- Select Categories, Models, Inference.
- And now just click Apply.

Step 2: Configure data source

Step 3: Configure data storage and processing

Step 4: Review and create

Select an embeddings model to convert your data into an embedding. Your selection may limit vector stores and embedding types that are available. Pricing depends on the model. [Learn more](#)

Titan Text Emb... v2.0 [Info](#) [Edit](#)
On-demand

Additional configurations

Vector store Info
Let Amazon create a vector store on your behalf or select a previously created store to allow Bedrock to store, update and manage embeddings. You will be billed directly from the vector store provider. [Learn more](#)

Vector store creation method

☒ **Quick create a new vector store - Recommended**
A vector store will be created on your behalf in this AWS account during Knowledge Base creation.

☐ Use an existing vector store
Connect to an existing vector store to store, update, and manage embeddings.

Vector store type - new [Info](#)
A vector store holds the vector embeddings representation of your data. This choice cannot be changed later. Not all vector stores support binary vector embeddings.

[Select a vector store](#)

[Cancel](#) [Previous](#) [Next](#)

- Choose Vector Store
 - Quick create a new vector store
- Choose Vector Store type as
 - Open Search Serverless
- And just click Next.

Step 3: Configure data storage and processing

Embeddings model

Model Titan Text Embeddings v2	Embedding type Float vector embeddings	Vector dimensions 1024
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Vector store

Quick create vector store - *recommended*
Amazon OpenSearch Serverless

Multimodal storage destination

S3 URI
-

Cancel Previous **Create Knowledge Base**

- Now just scroll down.
- Before Clicking Create Knowledge Base.
- Go to your created Bucket and store Some data like Pdfs.
- So go to your created S3 Bucket Tab

aws-assist-file

Objects (0)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
No objects				

You don't have any objects in this bucket.

Upload

- Now just click Upload.

Upload

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (0)

All files and folders in this table will be uploaded.

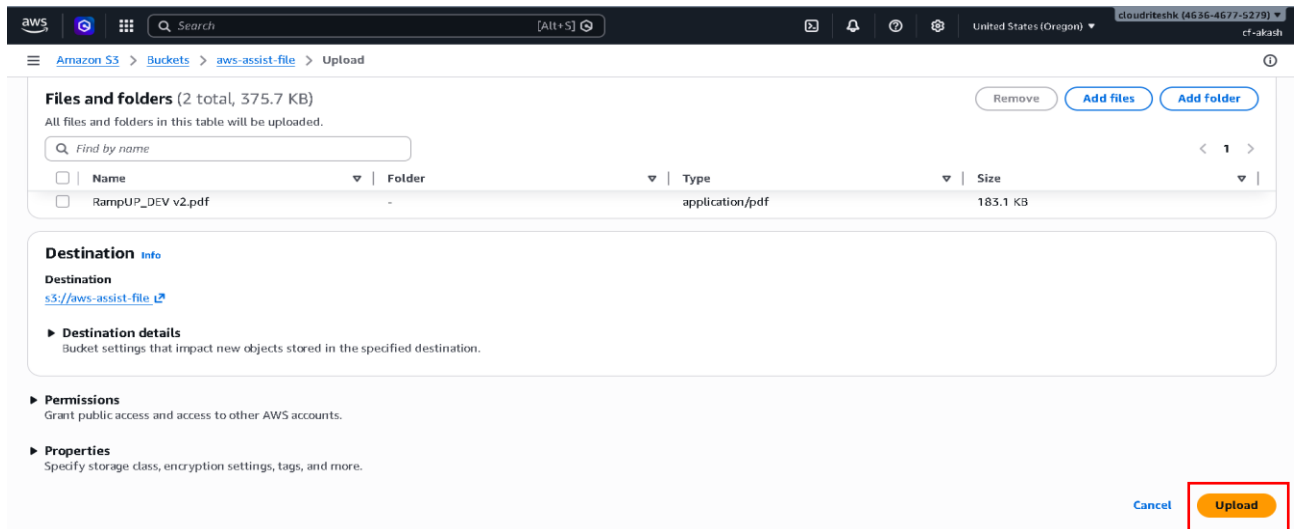
Find by name

Name	Folder	Type	Size
No files or folders			

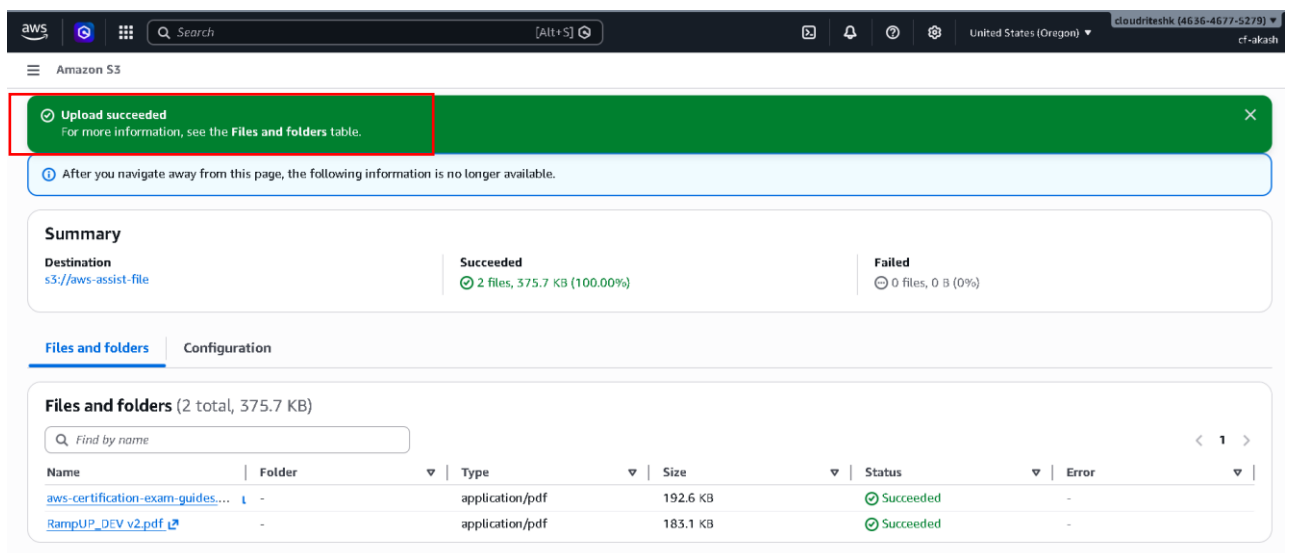
You have not chosen any files or folders to upload.

Remove **Add files** Add folder

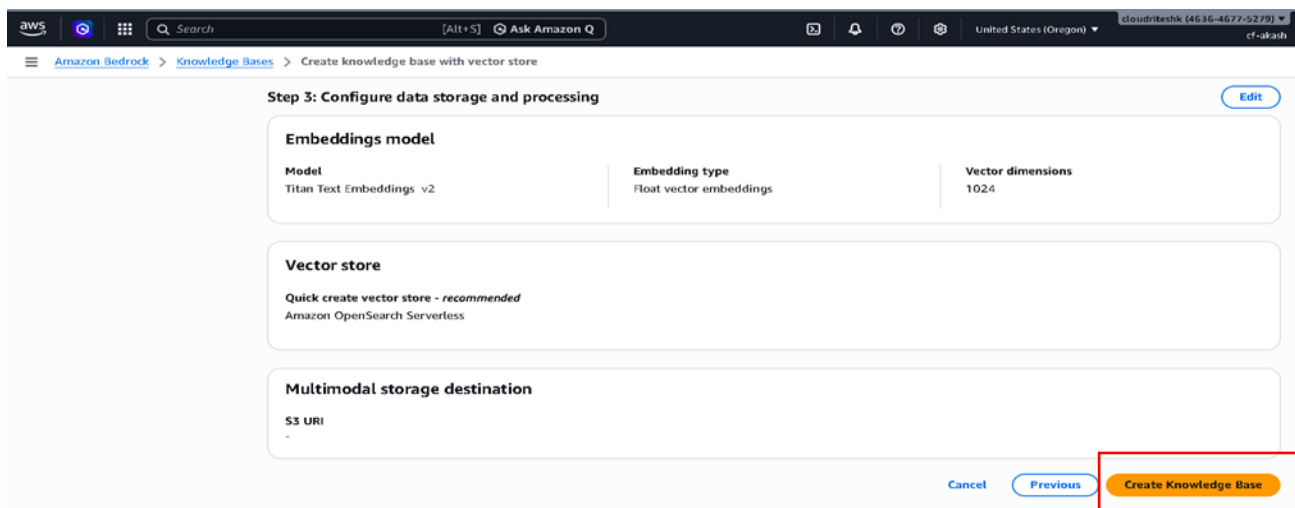
- Now click Add Files
- Select document that helps your agent in analyzing Data



- Now just scroll down and click upload.



- Now your Data is properly uploaded.
- Go Back to your Knowledge Base tab.



- Now just click Create Knowledge Base.

The screenshot shows the AWS console interface for the 'aus_asis_kb' Knowledge Base. A green notification bar at the top states: 'Knowledge Base 'aus_asis_kb' created successfully. Sync one or more data sources to index your content for searching. Syncing can take from a few minutes to a few hours.' Below this, the 'Knowledge Base overview' section displays the following details:

- Knowledge Base name:** aus_asis_kb
- Knowledge Base description:** -
- Service role:** AmazonBedrockExecutionRoleForKnowledgeBase_ceyy1
- Knowledge Base ID:** S3JELQSRBA
- Status:** Available
- Created date:** April 18, 2026, 13:41 (UTC+05:30)
- Log Deliveries:** Configure log deliveries and event logs in the [Edit](#) page.
- Retrieval-Augmented Generation (RAG) type:** Vector store

The 'Data source (1)' section shows a table with one data source:

Data source...	Status	Data source ty...	Account ID	Source Link	Last sync time	Last sync warn...	Text chunking ...	Parsing strategy
aws_agent_as...	Available	S3	46364677527...	s3://aws-assist...	-	-	Default	Default

- Our Knowledge Base is Created.

This screenshot shows the 'aus_asis_kb' Knowledge Base overview and the 'Data source (1)' section. The 'Sync' button in the 'Data source (1)' section is highlighted with a red box. The data source table is also highlighted with a red box:

Data source...	Status	Data source ty...	Account ID	Source Link	Last sync time	Last sync warn...	Text chunking ...	Parsing strategy
aws_agent_as...	Available	S3	46364677527...	s3://aws-assist...	-	-	Default	Default

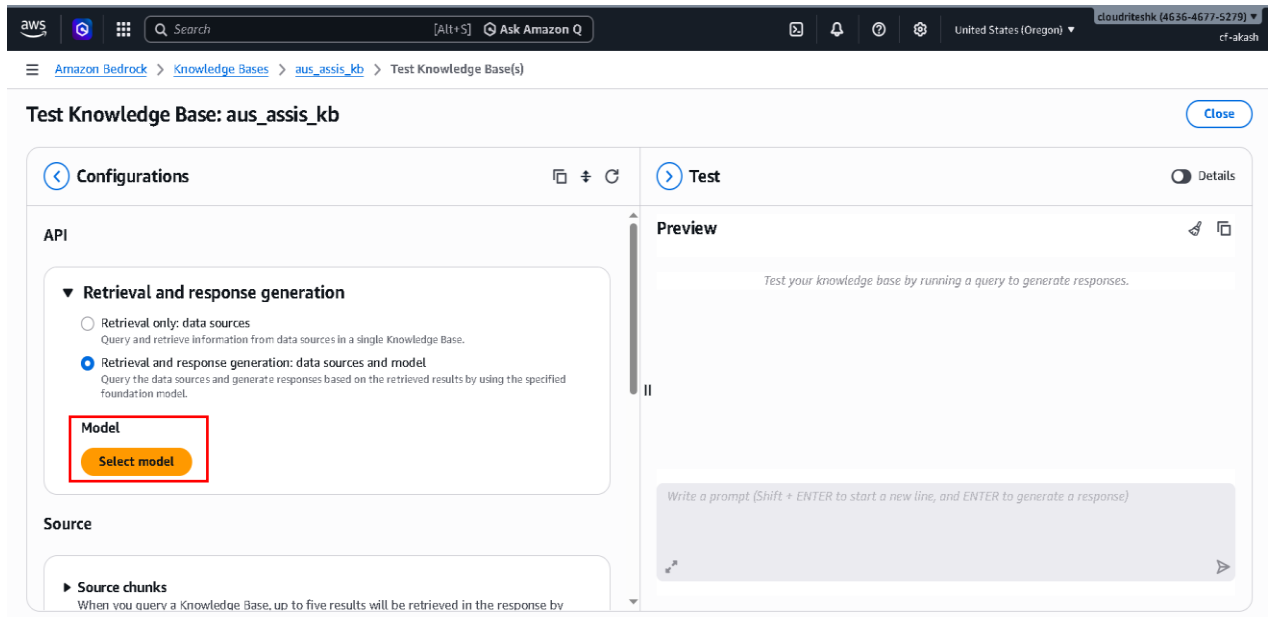
- Scroll Down
- But our Knowledge Base Data source is not Synced.
- Select data source and click Sync.
- It takes Time.

This screenshot shows the 'aus_asis_kb' Knowledge Base overview and the 'Data source (1)' section after the sync process is complete. A green notification bar at the top states: 'Sync completed for data source - 'aws_agent_assist''. The 'Test Knowledge Base' button in the 'Data source (1)' section is highlighted with a red box. The data source table is also highlighted with a red box:

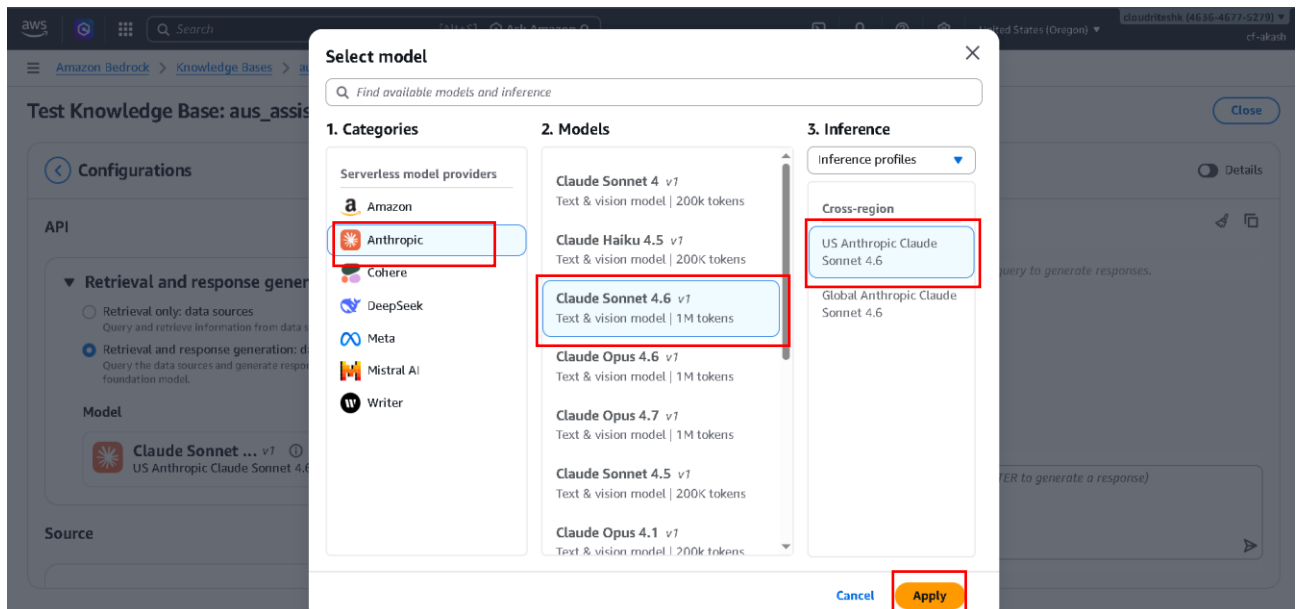
Data source...	Status	Data source ty...	Account ID	Source Link	Last sync time	Last sync warn...	Text chunking ...	Parsing strategy
aws_agent_as...	Available	S3	46364677527...	s3://aws-assist...	April 18, 2026,...	-	Default	Default

- Sync Completed.

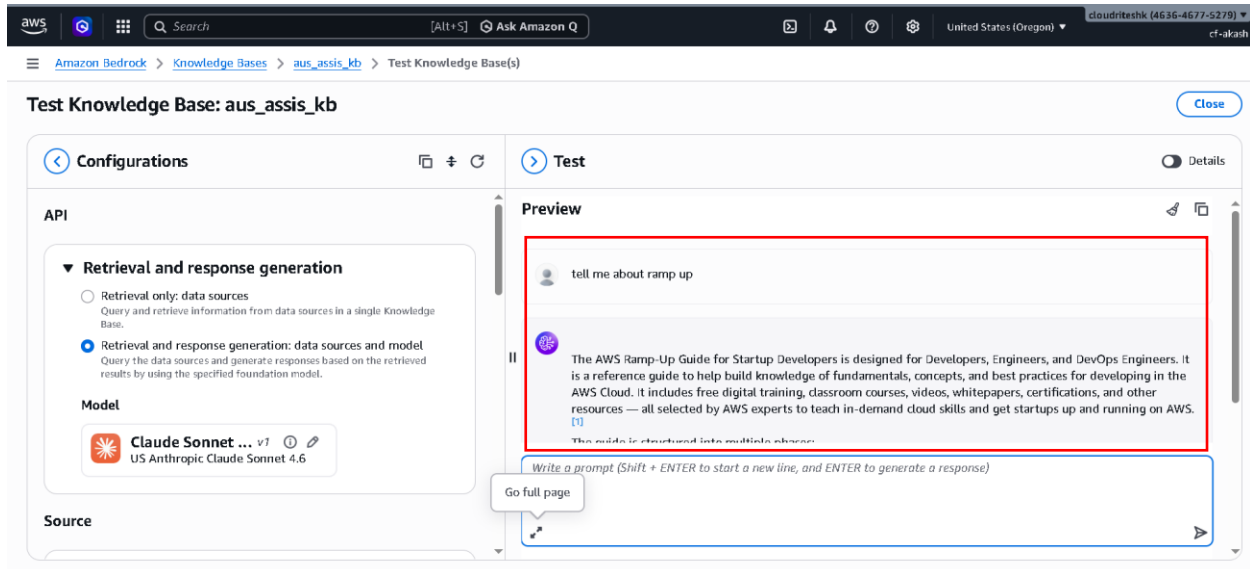
- Now just Click Test Knowledge Base.



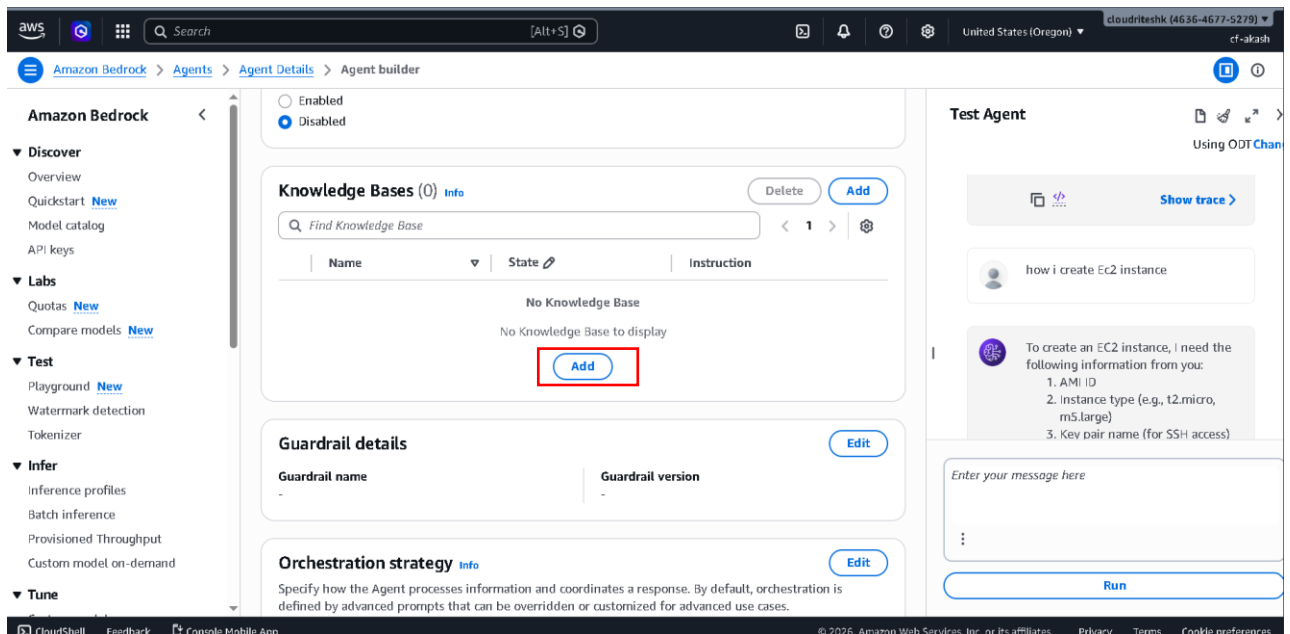
- Click Select Model.



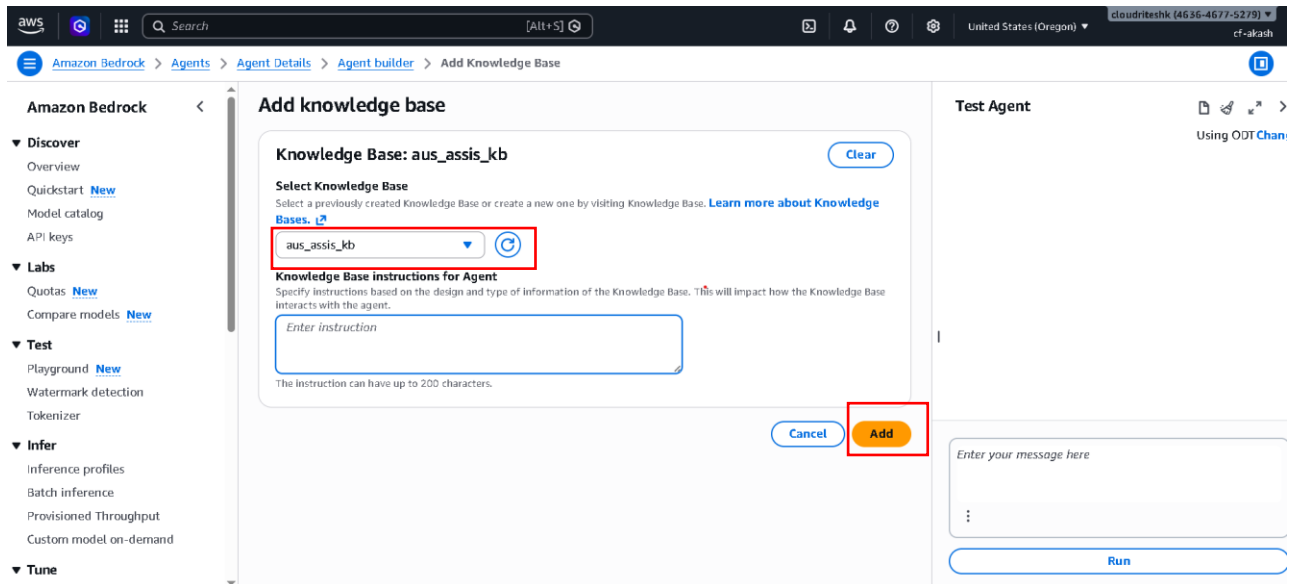
- Now Just Select Model Provider, Model and Inference
- And just click Apply.



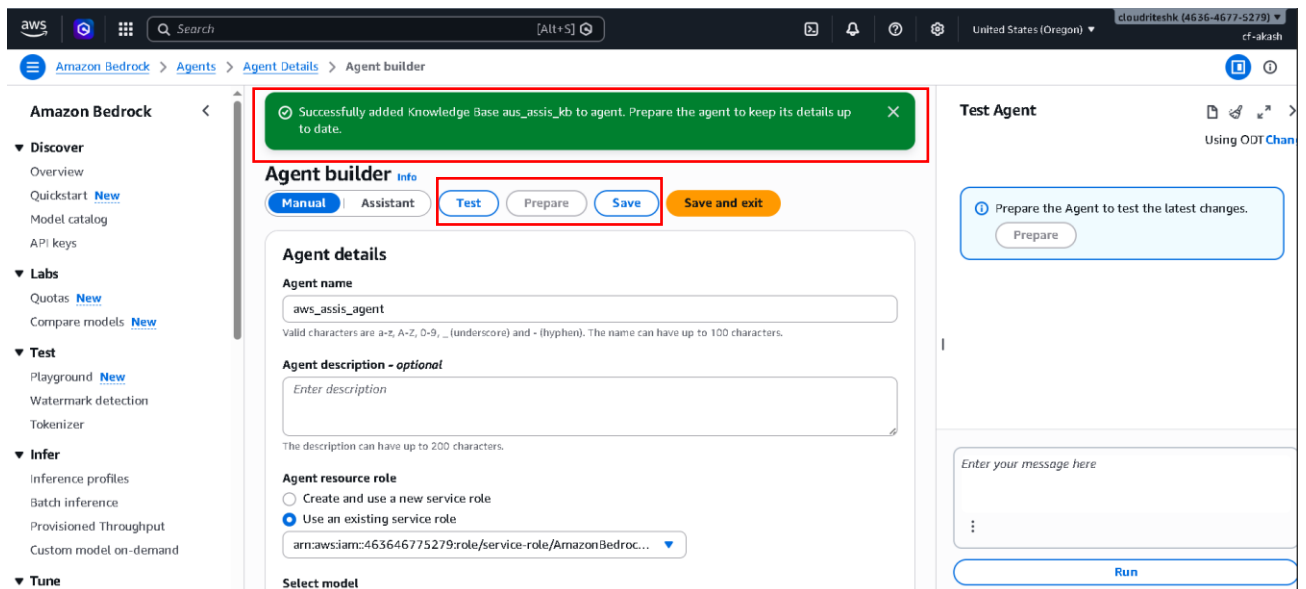
- Now test our Knowledge base
- Its Working Properly.
- Now go to your Agent tab.



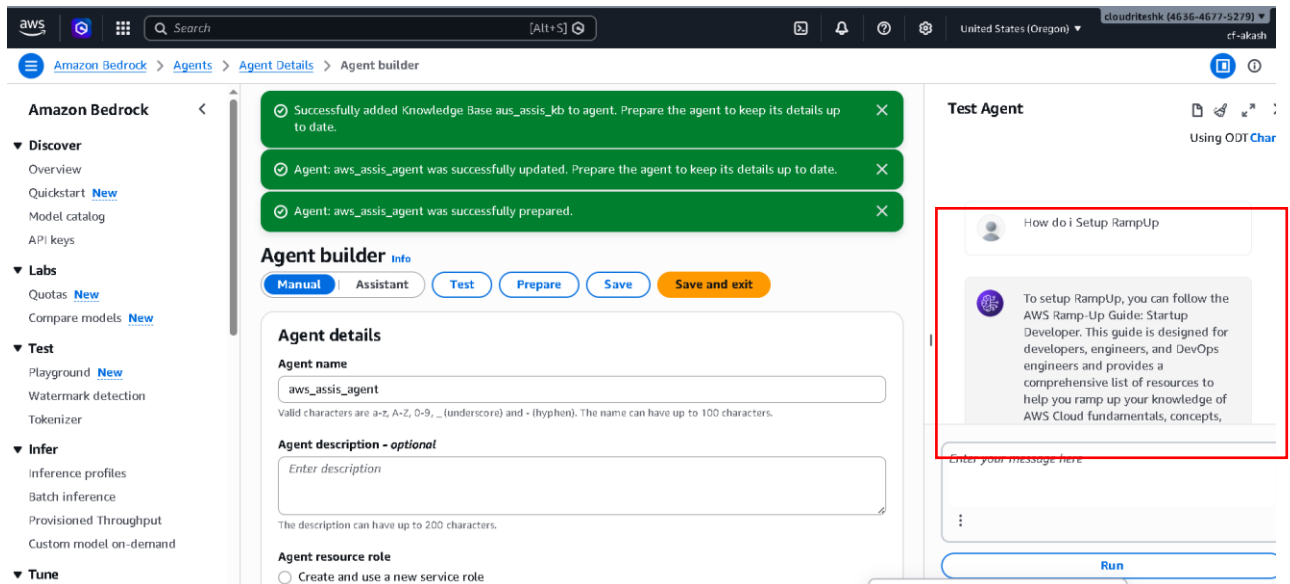
- And Scroll Down.
- Click Add inside Knowledge Bases.



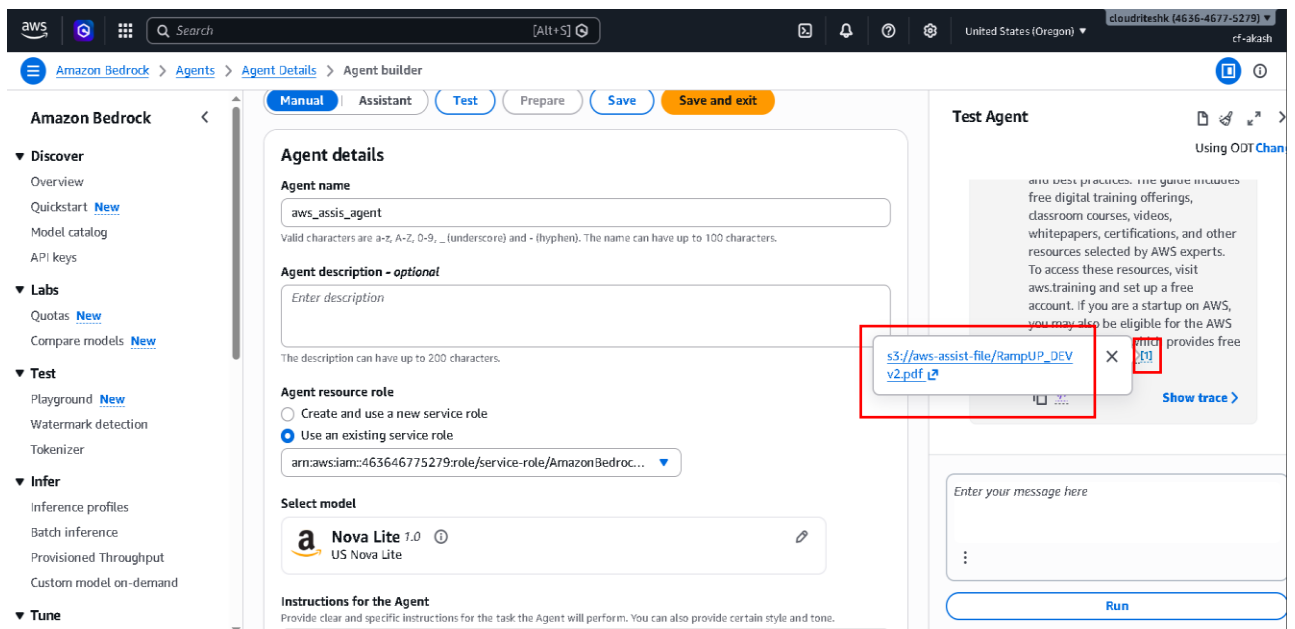
- Now Select your Knowledge Base.
- And click Add.



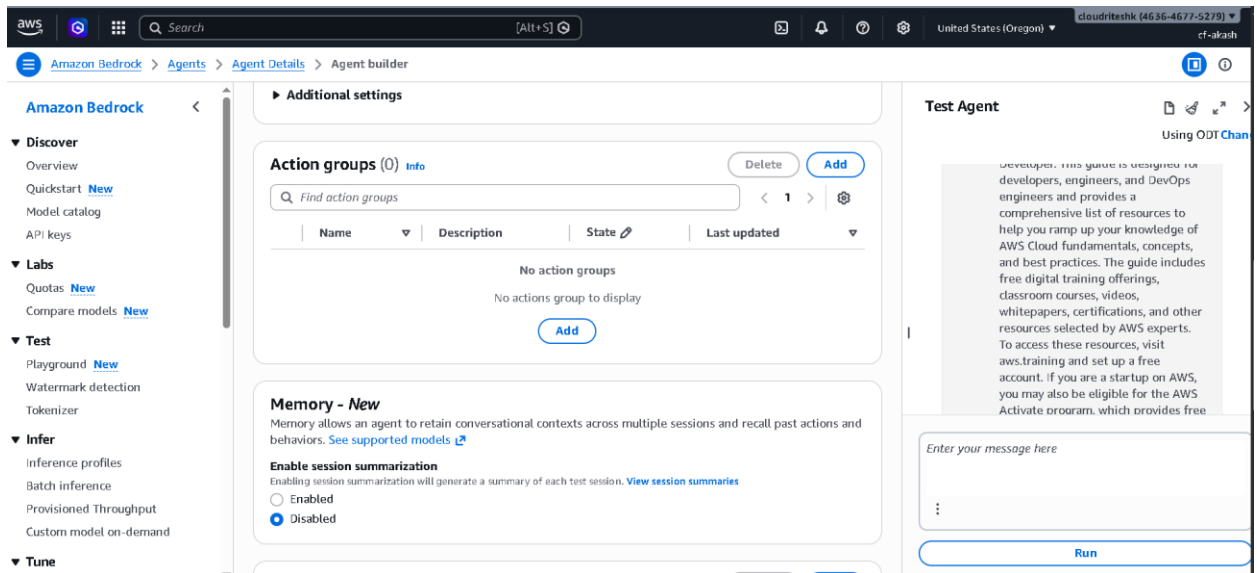
- Our Knowledge Base is Successfully added.
- Now click Save.
- Then Click Prepare.
- And then click Test.
- Now Ask Some Question.



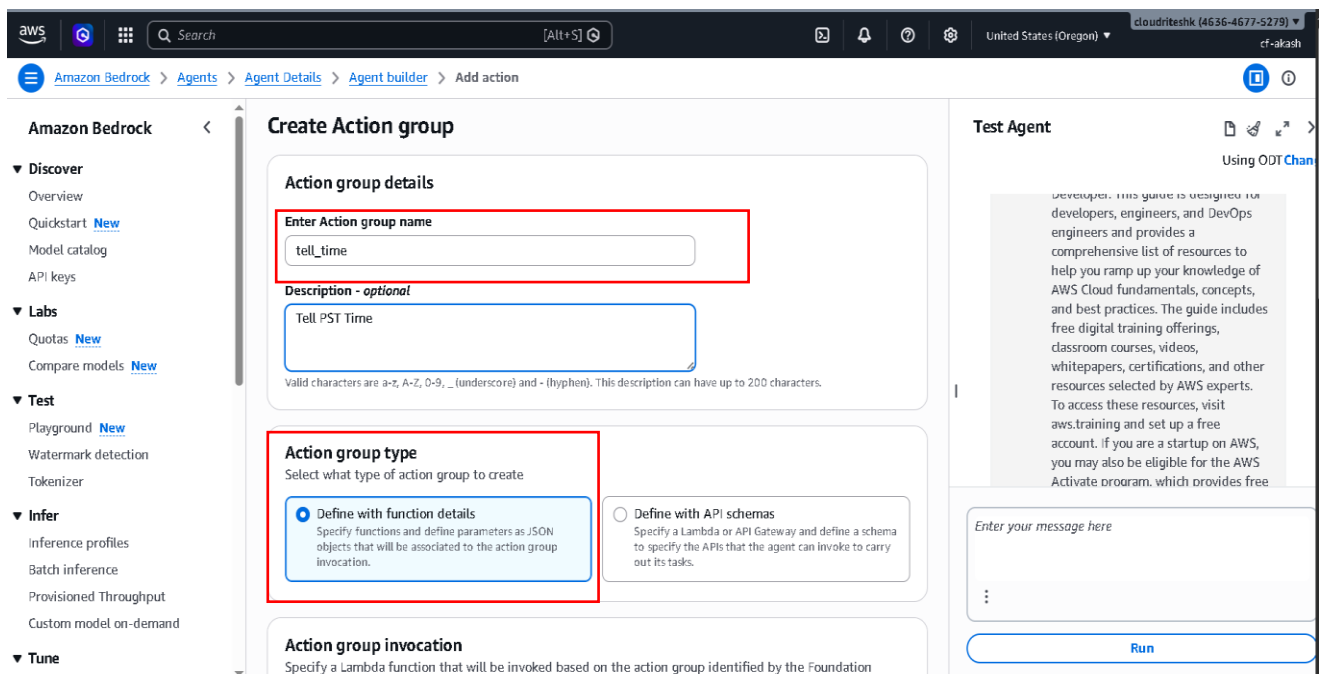
- Now Our Agent is Properly Working.
- Now scroll Down and Check What is the Source of the Output.



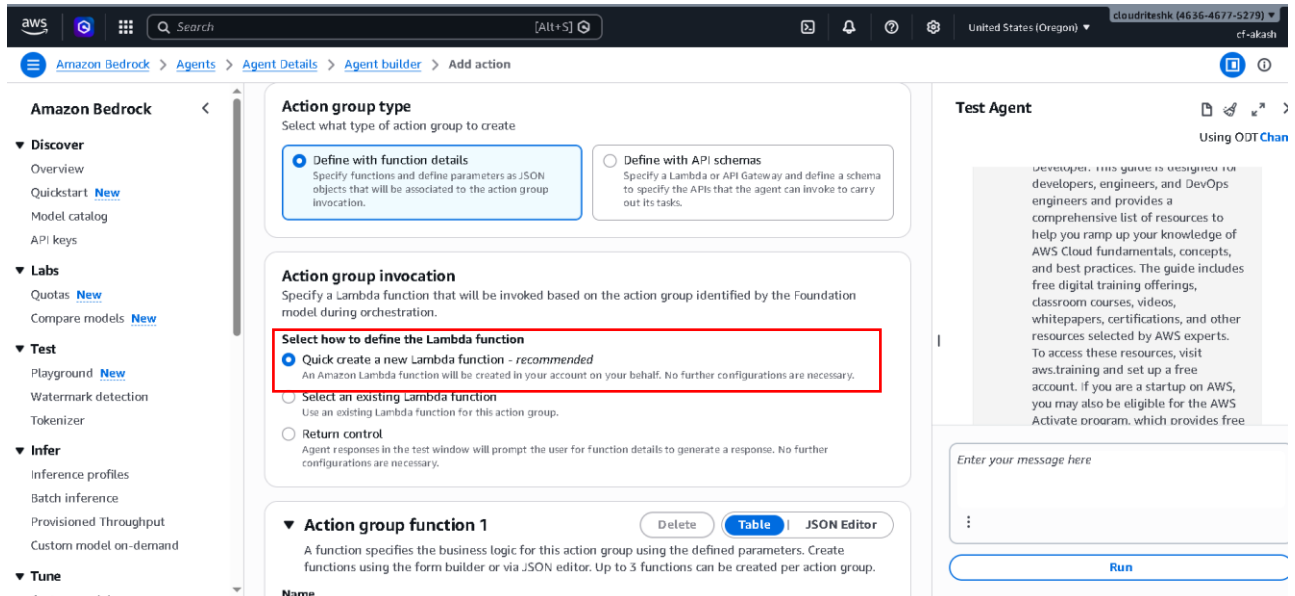
- Now when you click 1.
- You see the source and the source is from your Data Source that you uploaded.
- Now We add Action Groups in Our Agent.
- Well it means we can actually pass in piece of function or other things functionalities that we can add into this agent workflow



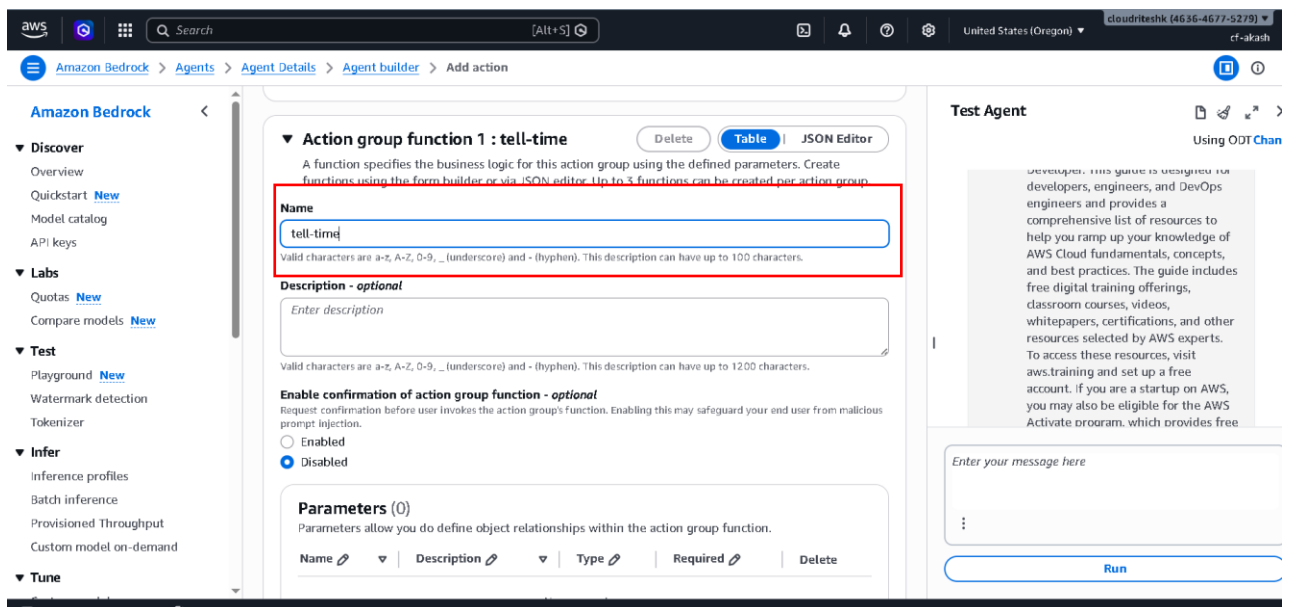
- So now just scroll Down
- And click Add



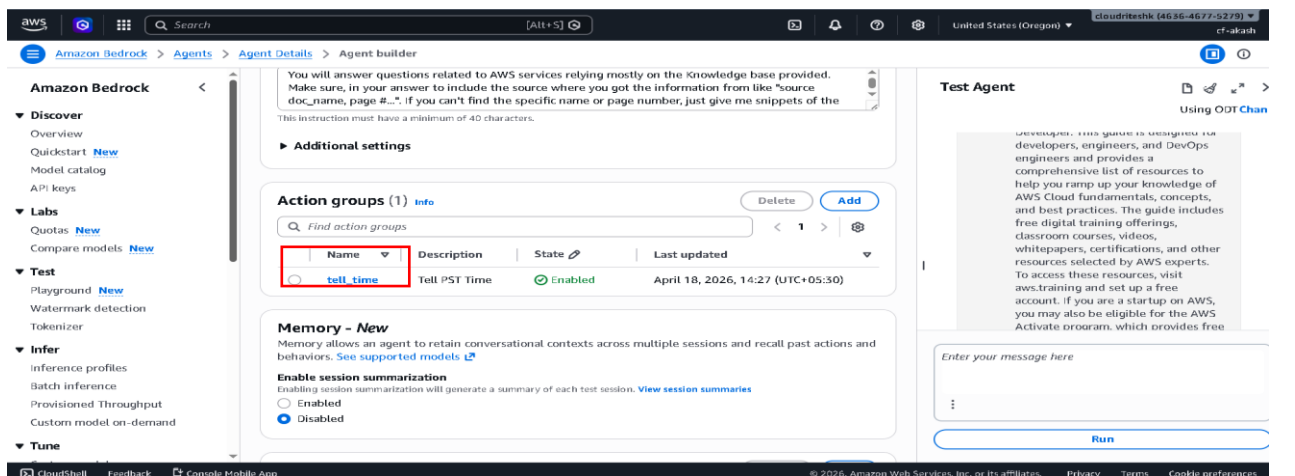
- Enter Action Group name.
- Action Group Type
 - Define with function Details



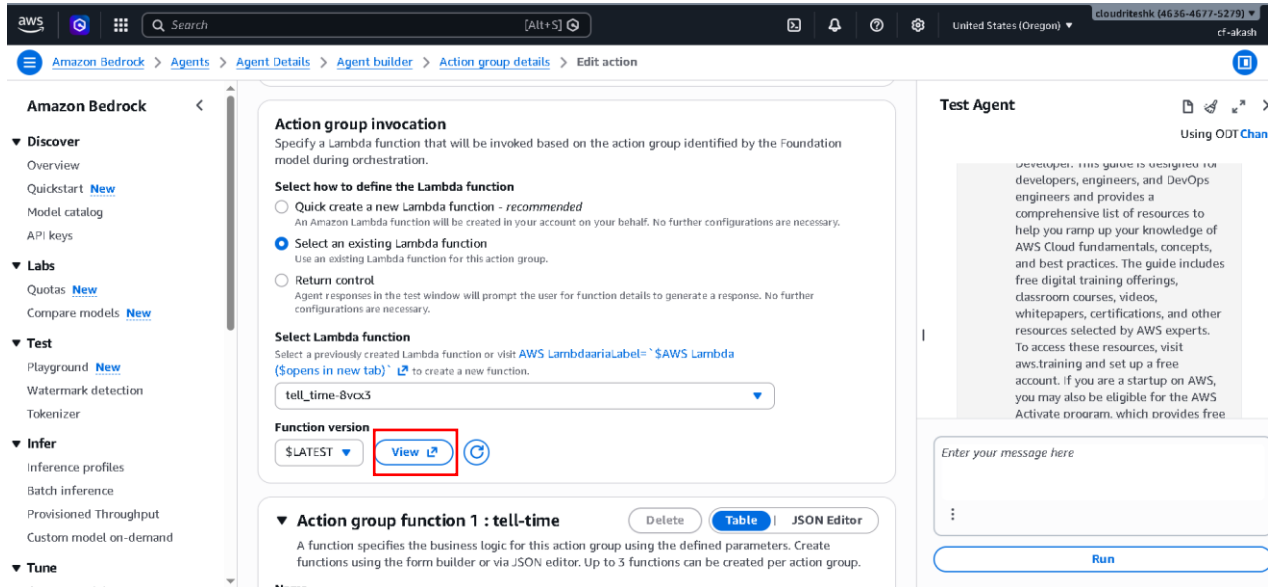
- Action Group invocation
 - Quick create a new Lambda Function



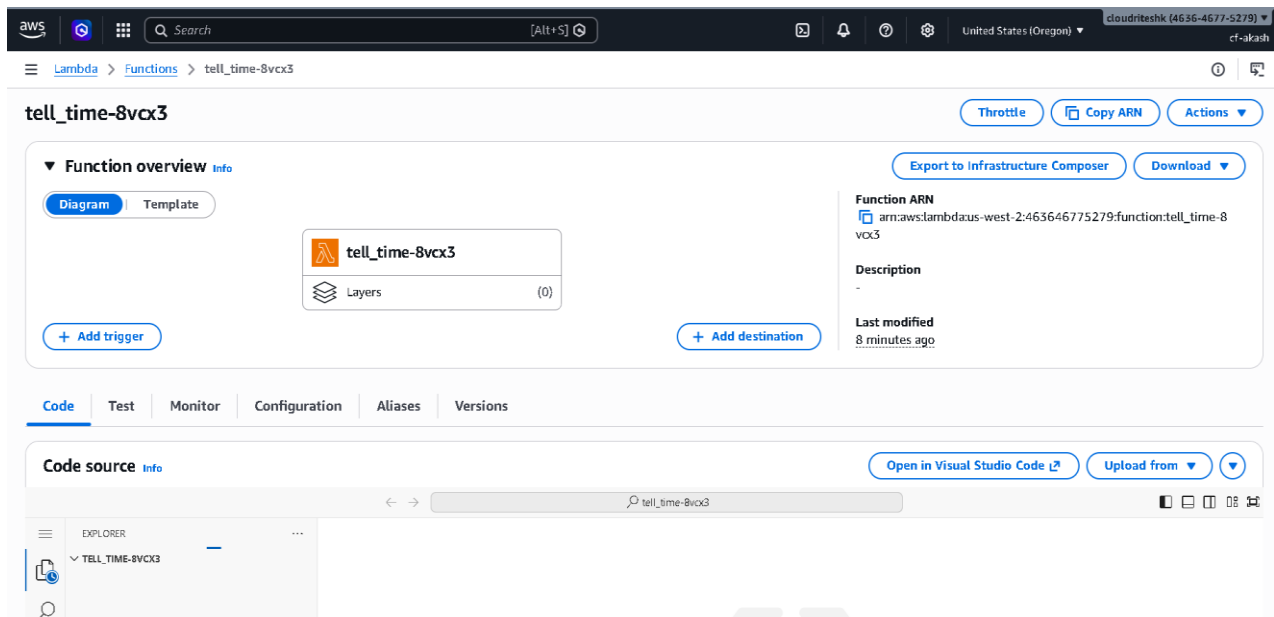
- Now give A group function1 name.
- And just scroll Down and click Create.



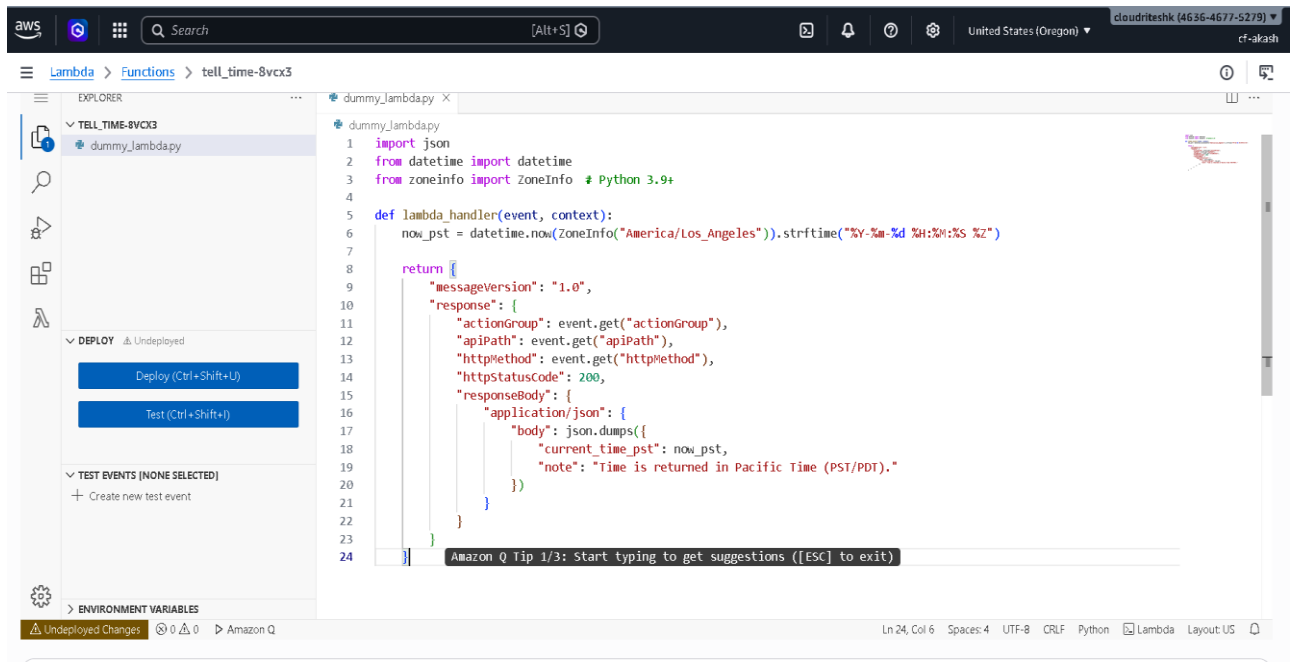
- Now see Our Action Group is Created.
- Click on it and go inside.



- Now just scroll Down.
- And inside Action Group invocation click View.



- Now you can see one lambda function is created.
- Scroll down.



- Now Remove Previous Written code and Paste this Code.

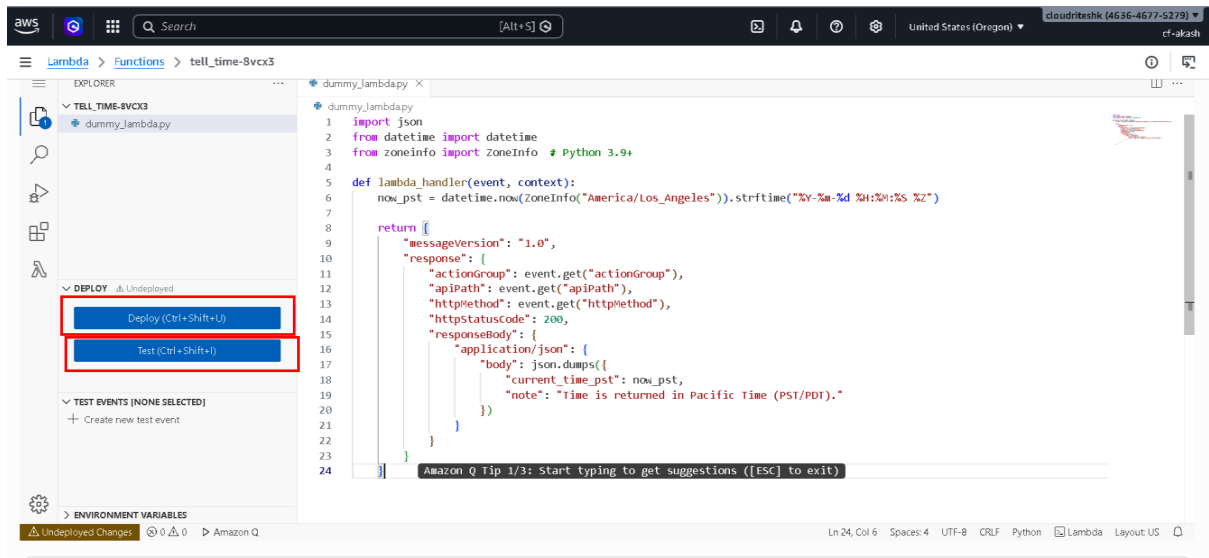
```

import json
from datetime import datetime
from zoneinfo import ZoneInfo # Python 3.9+

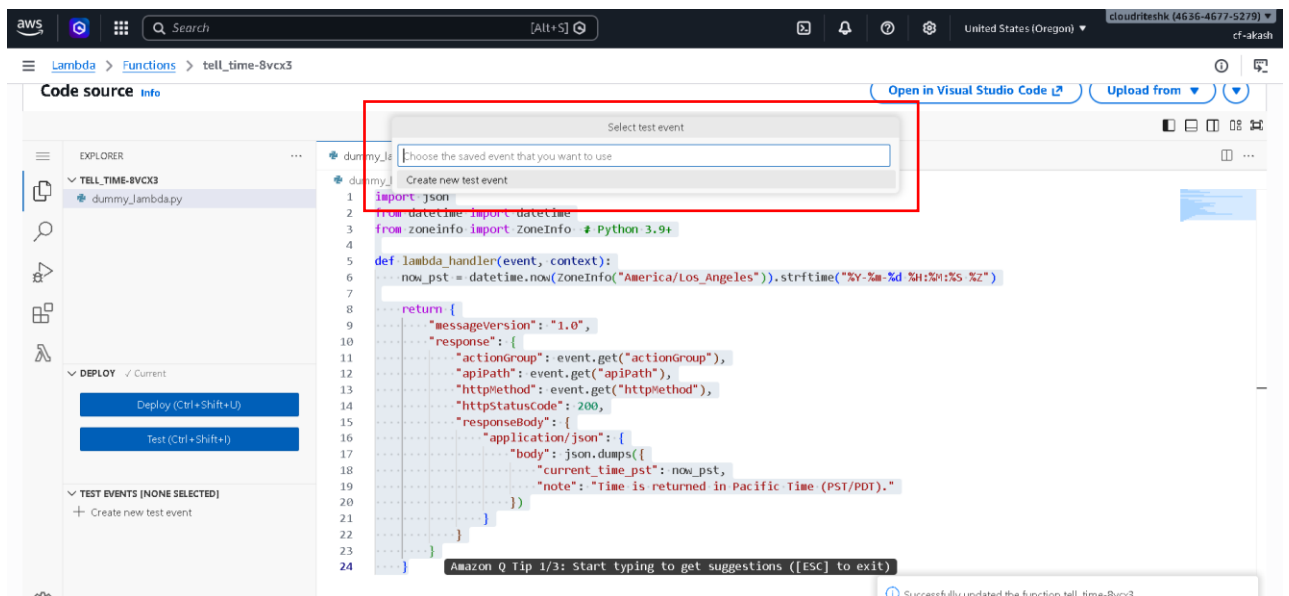
def lambda_handler(event, context):
    now_pst = datetime.now(ZoneInfo("America/Los_Angeles")).strftime("%Y-%m-%d
%H:%M:%S %Z")

    return {
        "messageVersion": "1.0",
        "response": {
            "actionGroup": event.get("actionGroup"),
            "apiPath": event.get("apiPath"),
            "httpMethod": event.get("httpMethod"),
            "statusCode": 200,
            "responseBody": {
                "application/json": {
                    "body": json.dumps({
                        "current_time_pst": now_pst,
                        "note": "Time is returned in Pacific Time (PST/PDT).",
                    })
                }
            }
        }
    }

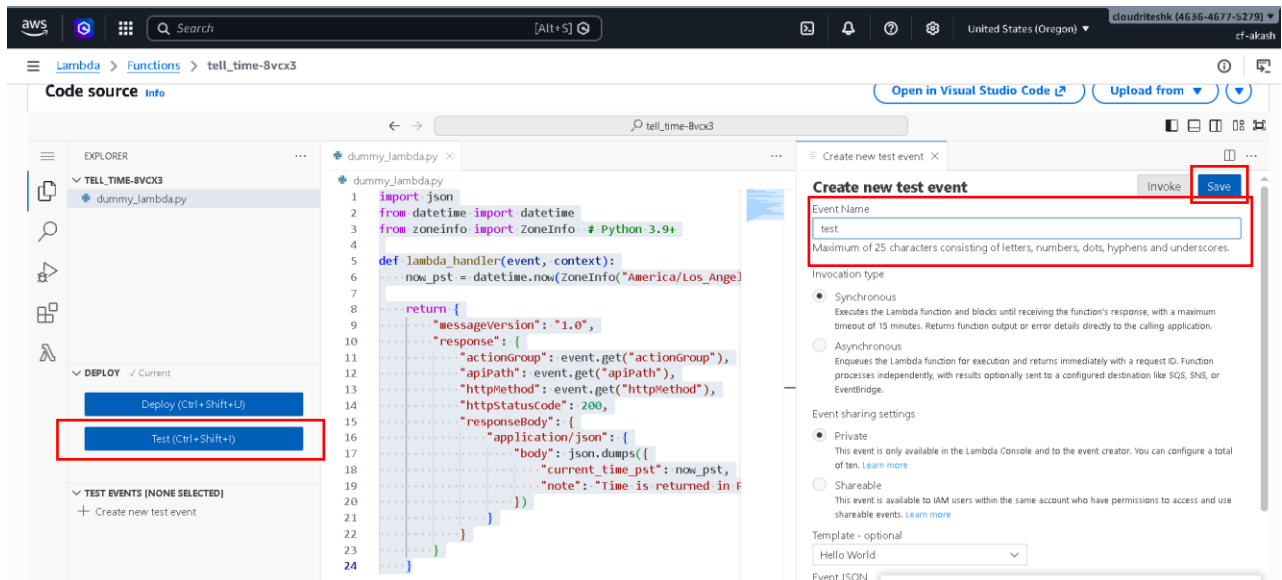
```



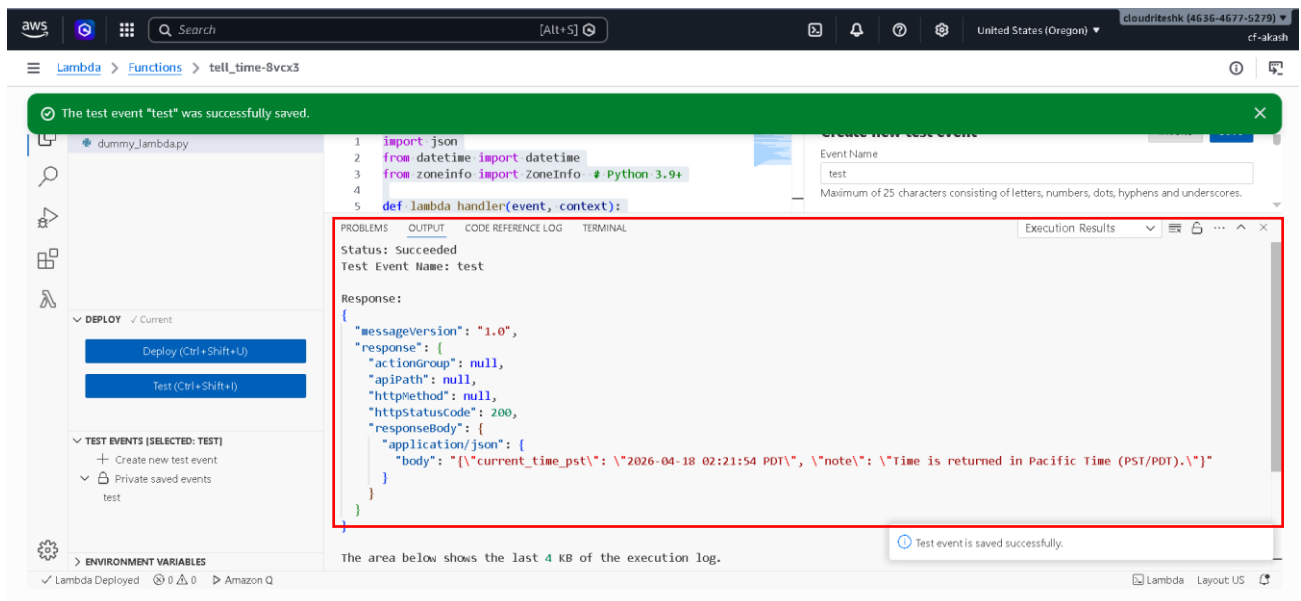
- Now Click Deploy.
- And then code is deployed click Test.



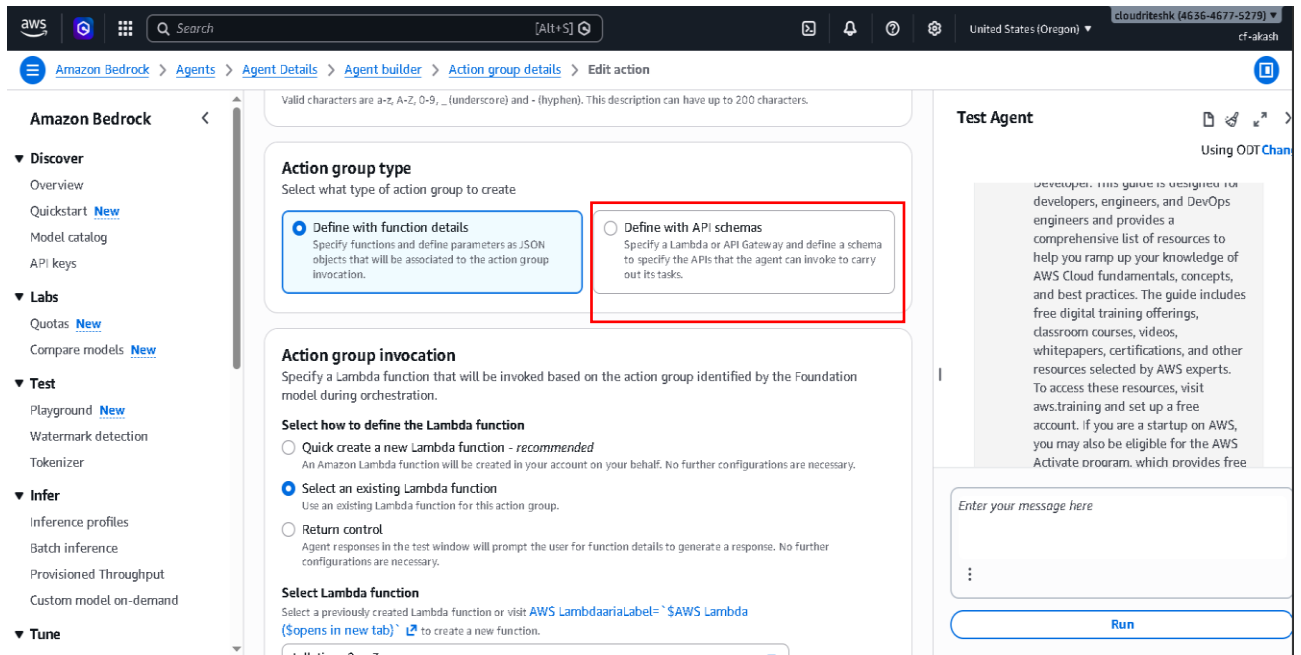
- Now click Create New Test Event.



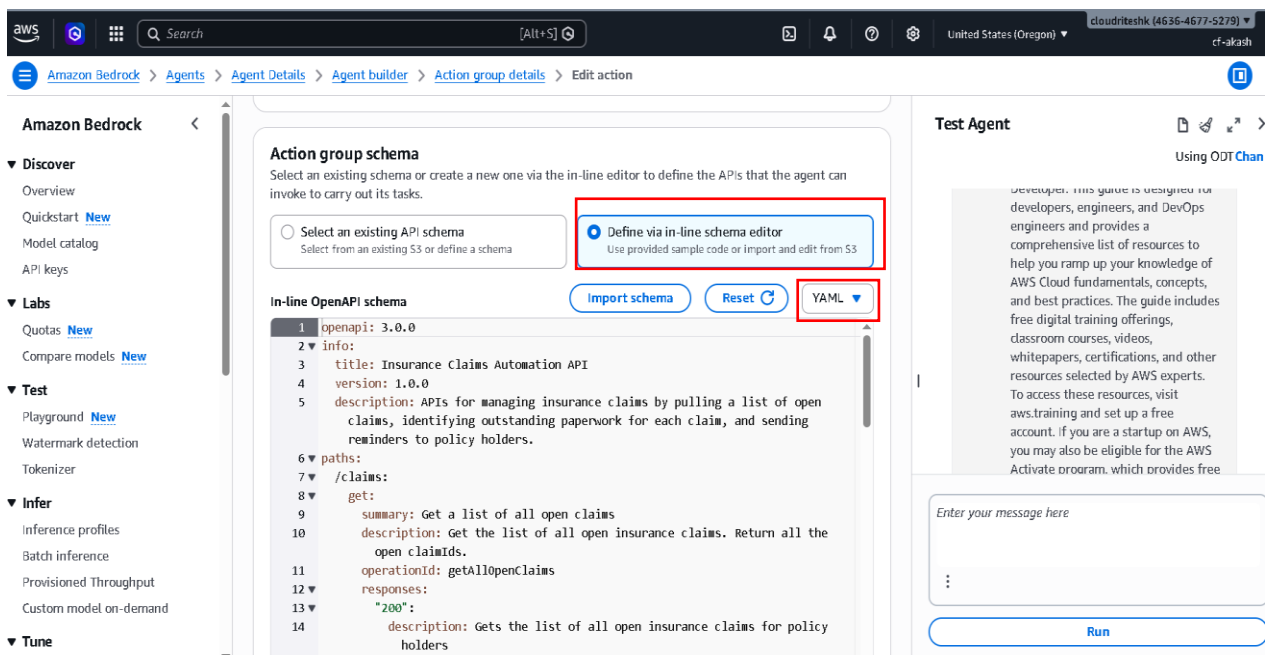
- Write Event Name.
- And then click Save.
- And again click Test.



- Our Test case is running
- Now what I need to do here I need to make sure that I pass a little bit more information so that we can map things between the LLM the agent and tell it that okay, when you get Question related to time you need to look into this structure which will allow the LLM to actually infer from the prompt that may receive to call this function this tool



- Go back to your Agent tab
- And scroll up.
- And Choose Action Group type
 - Define with API schema



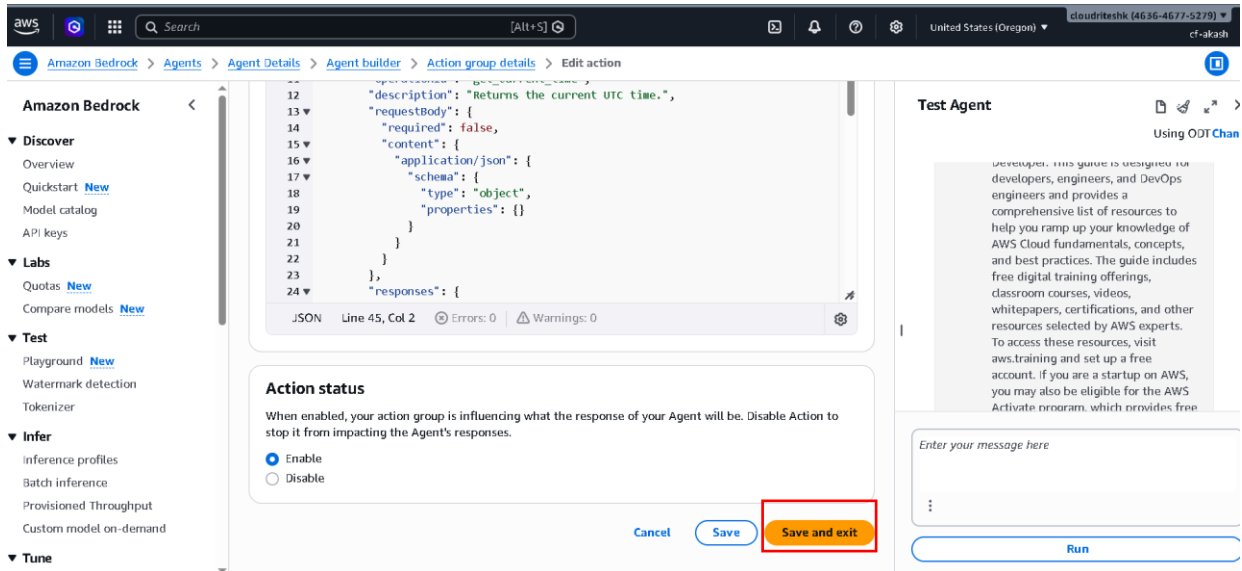
- Now scroll Down And in Action Group Schema choose
 - Define via in-line editor
- Then Change Schema to Jason.
- Now Remove the Previous Written inline Schema and paste this

```
{
  "openapi": "3.0.0",
  "info": {
    "title": "Time API",
    "version": "1.0.0",
  }
}
```

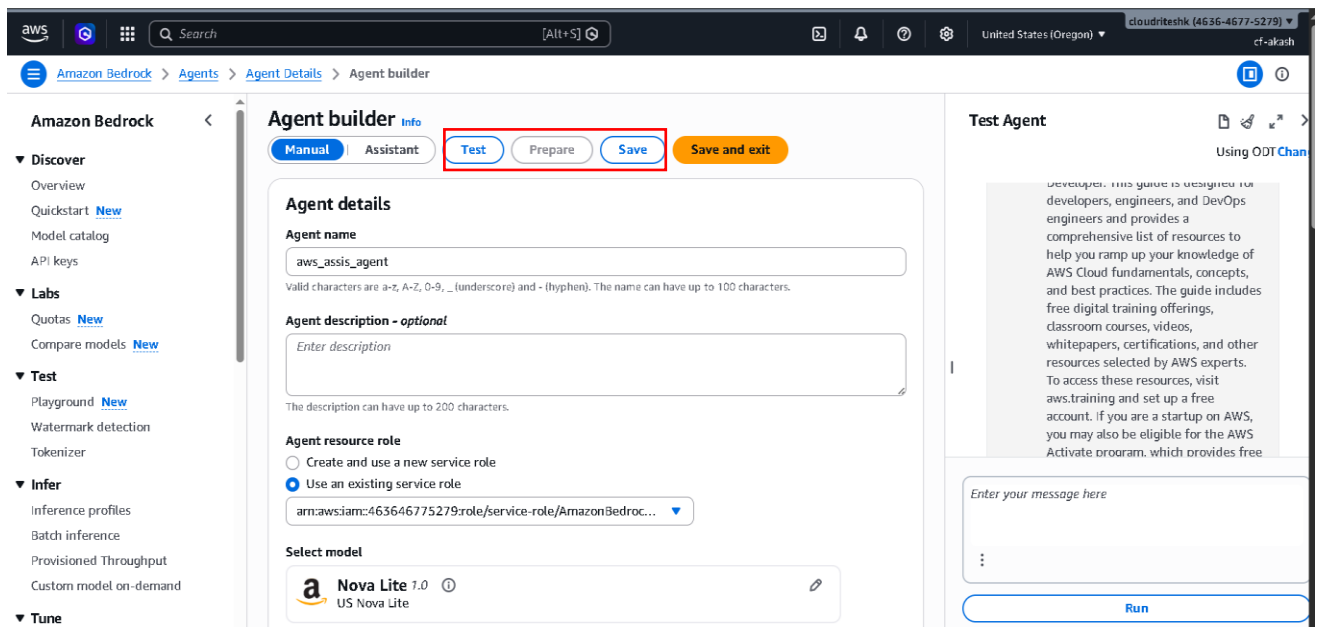
```

    "description": "API to get the current UTC time."
  },
  "paths": {
    "/get-time": {
      "post": {
        "operationId": "get_current_time",
        "description": "Returns the current UTC time.",
        "requestBody": {
          "required": false,
          "content": {
            "application/json": {
              "schema": {
                "type": "object",
                "properties": {}
              }
            }
          }
        },
        "responses": {
          "200": {
            "description": "Successful response containing the current UTC
time.",
            "content": {
              "application/json": {
                "schema": {
                  "type": "object",
                  "properties": {
                    "current_time": {
                      "type": "string",
                      "description": "The current time in UTC."
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}

```



- Now Scroll Down and just click Save and Exit.



- Now just click save, Then Prepare and Test.
- Now ask Question Like tell Me time.

aws [Search] [Alt+S] United States (Oregon) cloudriteshk (4636-4677-5279) cf-akash

Amazon Bedrock > Agents > Agent Details > Agent builder

Amazon Bedrock

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Manual Assistant **Test** Prepare Save Save and exit

Agent details

Agent name

aws_assis_agent

Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 100 characters.

Agent description - optional

Enter description

The description can have up to 200 characters.

Agent resource role

☐ Create and use a new service role

☒ Use an existing service role

arn:awsiam:463646775279:role/service-role/AmazonBedroc...

Test Agent

Using ODT Chan

Tell me the time

The current time is 2026-04-18 02:46:33 PDT.

Show trace >

Enter your message here

Run

- Now see its Properly working.